Unit 3 Overview: Lines and Angles

Unit Narrative

This unit begins with recalling the vocabulary and angle pair relationships explored in previous courses, and how these relationships are used in solving for variables and angle measures. These relationships become the foundation used to discover the angle pair relationships formed when parallel lines are intersected by a transversal. Students prove these relationships formally through two-column, paragraph, and flow cart proofs to develop theorems. The relationships discovered are used to solve mathematical and real-world problems.

Section 1 of this unit focuses on discovering relationships between various angle pairs. In Lessons 1–3, students use copies of angles and transformations to determine the angle relationships of various angle pairs. In Lesson 1, students revisit congruent, supplementary, and vertical angles from previous courses, and use these relationships to find missing angle measures. Formal theorems and corollaries are stated, defining these angle relationships. Lesson 2 introduces the vocabulary of a transversal and the relationship between corresponding angles when the lines being intersected by a transversal are parallel. Students build on the relationship of corresponding angles to discover and prove that alternate interior and alternate exterior angles are congruent when parallel lines are intersected by a transversal in Lesson 3, and consecutive interior and exterior angles are supplementary in Lesson 4. Lesson 5 focuses on applying the postulates and theorems learned in Lessons 1–4 to determine if two lines are parallel. Students use what they learned in Unit 2 as they write the converse of the postulates and theorems about parallel lines to determine if the lines are parallel.

Section 2 begins with students applying all the angle pair relationships formed by parallel lines and transversals to solve mathematical and real-world problems by determining measures of unknown angles and the value of variables in Lesson 6. Students work together on a Card Sort activity and Collaboration in Lesson 7 to demonstrate their knowledge of these angle pair relationships.

Section 3 focuses on constructing proofs using angle pair relationships formed by parallel lines and transversals. Instruction includes constructing or completing two-column, flow chart, and paragraph proofs. The definitions, properties, postulates, and theorems learned in Lessons 1-7 are used as justification in the steps needed to prove a conclusion from a given set of information.

Unit At-A-Glance

Section 1: Proving Angle Pair Relationships of Parallel Lines Crossed by a Transversal	Benchmark Addressed	Lesson 1	Proving Angle Relations
	MA.912.GR.1.1 **MA.912.LT.4.10** **MA.912.LT.4.3**	Lesson 2	Vertical and Corresponding Angles
		Lesson 3	Congruent Angles in Parallel Lines
		Lesson 4	Consecutive Angles
		Lesson 5	Converse of the Alternate Interior Angles Theorem
Section 2: Applying Angle Pair Relationships to Solve Problems	Benchmark Addressed	Lesson 6	Using Angle Relationships, Postulates, and Theorems
	MA.912.GR.1.1	Lesson 7	Solving Problems Using Angle Pair Relationships
Section 3: Proofs Using Angle Pair Relationships of Parallel Lines and Transversals	Benchmark Addressed	Lesson 8	Angle Relationship Proofs – Part 1
	MA.912.GR.1.1 **MA.912.LT.4.10**	Lesson 9	Angle Relationship Proofs – Part 2

Differentiation

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Engagement
The "WHY" of learning

- Allow multiple opportunities for students to use the Think-Pair-Share routine as they develop their ideas, so that it fosters collaboration and builds a mathematical community among students.
- Consider implementing a silent board game when completing steps in a proof. Project the proof and allow students to choose where to engage by silently coming to the board to complete one statement or reason that they feel comfortable with providing a response for. Allowing this choice and rewarding participation will recruit interest in the often challenging task of writing proofs.
- The Wrap-Ups in Lessons 8 and 9 provide students the opportunity to self-assess and reflect as the unit concludes.

Fill in the table to complete the proof.

Statement	Reason
1.	1. Given
2.	2. Vertical angles are congruent.
3.	3.
4.	4. Transitive Property of Congruence

Representation
The "WHAT" of learning

- Providing Word Walls and word banks for students throughout these lessons clarifies the key terms and supports decoding of the notation and symbols used as they continue to develop their understandings in the course.
- Consider having students create a Proof Resource in the form of a booklet or Foldable. Students could include definitions, properties, postulates, and theorems with examples and pictures. Examples of statements that "typically" follow each other in a proof could also be included. See the image for an example of statements that often follow each other.

$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	Given
$\angle GNB$ is a right angle	Definition of Perpendicular Lines
$m\angle GNB = 90°$	Definition of Right Angle

Action & Expression
The "HOW" of learning

- Allow students to use multiple tools such as patty paper, straightedges, and compasses to construct angles and to prove that they are congruent as often as they need in Lessons 1–4. Using such tools helps foster deeper understanding as students explore these angle relationships.
- Lesson 7 provides students the opportunity to enhance their capacity for self-monitoring their own progress on mastering the lesson learning targets by reflecting on their understanding of each on a rating scale.

1. I can use corresponding angles to determine angle measurements.

   I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this, and I feel confident.

2. I can use alternate interior angles to determine angle measurements.

   I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this, and I feel confident.

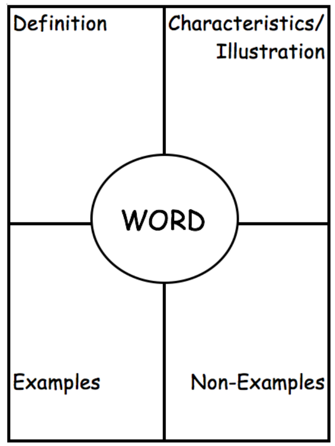
3. I can use alternate exterior angles to determine angle measurements.

   I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this, and I feel confident.

ELL Information

Students will be identifying and interpreting multiple angle relationships, definitions, and theorems. Consider providing the following supports for ELL students:

- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. For this section, students will be introduced to many concepts that set the foundation for writing proofs, so using a resource such as this could set students up to have points of reference in future applications.
- Word banks/Word Walls in lessons where vocabulary is introduced. Also consider providing an image/visual that represents the word so it is easier for students to make that connection.
- Word bank when completing the proofs in Lessons 8 and 9 with the terms and reasoning that ELL students can use when completing proofs.



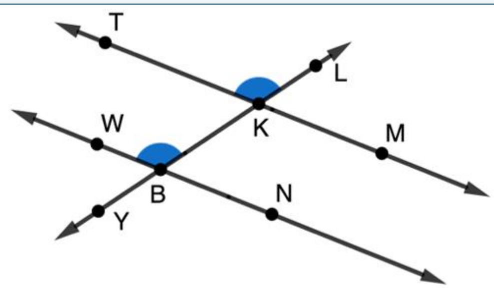
All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated
Support" below:

- Complementary angles
- Corresponding angles
- Supplementary angles
- Transversal
- Vertical angles

Additional Differentiated Support

Scaffolding Resources

Highlighting Text can be used to help students approach more complicated and wordy problems or proofs in all the lessons. Provide students with highlighters or colored pencils for them to label and highlight angles in problems so that they can be more easily identified. This will also help with students who are more visual learners and have a difficult time visualizing angles and their relationship with other angles.



Patty Paper can be used as a support for students who need it when determining if angles are congruent.

- While patty paper is used in Lessons 1–3 as an activity to prove angles are congruent, consider having this tool available for students as an additional support for the remainder of the lessons, and let students know that it is there if they need it.

On-Ramp

Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Lines, Line Segments, and Rays
- Parallel and Perpendicular Lines
- Right, Acute, and Obtuse Angles

Section Overview

Section 1: Proving Angle Pair Relationships of Parallel Lines Crossed by a Transversal (Lessons 1-5)

Benchmarks Addressed

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

- *Clarification 1: Postulates, relationships and theorems include vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

MA.912.LT.4.3 Identify and accurately interpret "if…then," "if and only if," "all" and "not" statements. Find the converse, inverse and contrapositive of a statement.

- *Clarification 1: Instruction focuses on recognizing the relationships between an "if…then" statement and the converse, inverse and contrapositive of that statement.*
- *Clarification 2: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

Learning Targets

Lesson	Learning Targets
Lesson 1	• I can solve problems that use the relationships of congruent supplements, congruent complements, and vertical angles. • I can use the relationships of congruent supplements, congruent complements, and vertical angles in proofs.
Lesson 2	• I can use vertical angles to solve problems. • I can use corresponding angles formed by parallel lines and a transversal to solve problems.
Lesson 3	• I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.
Lesson 4	• I can determine the relationships of consecutive exterior and interior angles. • I can solve problems that involve angles formed by parallel lines intersected by a transversal.
Lesson 5	• I can write and use the converse of the postulate or theorem. • I can use postulates and theorems about parallel lines to determine if lines are parallel.

Section 2: Applying Angle Pair Relationships to Solve Problems (Lessons 6-7)		
Benchmarks Addressed	Learning Targets	
MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles. • *Clarification 1: Postulates, relationships and theorems include vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.* • *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.* • *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*	Lesson 6	• I can solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.
	Lesson 7	• I can solve mathematical and real-world problems involving angle pair relationships formed by transversals and parallel lines.

Section 3: Proofs Using Angle Pair Relationships of Parallel Lines and Transversals (Lessons 8-9)		
Benchmarks Addressed	Learning Targets	
MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles. • *Clarification 1: Postulates, relationships and theorems include vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.* • *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.* • *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.* **MA.912.LT.4.10** Judge the validity of arguments and give counterexamples to disprove statements. • *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*	Lesson 8	• I can prove relationships using parallel lines and their transversals.
	Lesson 9	• I can prove relationships using parallel lines and their transversals.

Vertical Alignment

Grade 8 Unit 13: Properties and Theorems of Angles → **Geometry Unit 3 Lines and Angles**

Prior Course(s)	Course Level
MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	**MA.912.GR.1.1** Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. **MA.912.LT.4.10** Judge the validity of arguments and give counterexamples to disprove statements. **MA.912.LT.4.3** Identify and accurately interpret “if…then,” “if and only if,” “all,” and “not” statements. Find the converse, inverse, and contrapositive of a statement.

3.1 Proving Angle Relations

Lesson Introduction

Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can solve problems that use the relationships of congruent supplements, congruent complements, and vertical angles. • I can use the relationships of congruent supplements, congruent complements, and vertical angles in proofs.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How can you use angle relationships to identify congruent complements? • How can you use angle relationships to identify congruent supplements? • How can you use the linear pair postulate to identify congruent angles? • What relationship do vertical angles have?
Lesson Narrative	This lesson reviews the definitions of supplementary, complementary, and vertical angles from prior courses and uses these angle relationships to determine others. This lesson includes completing paragraph and two-column proofs to prove the relationships through the transitive and substitution properties. Students use the congruent supplements theorem to show linear pairs are congruent and identify a pair of vertical angles.

Component Summary				
	3.1.1 Warm-Up (~5 min.)	Which One Doesn't Belong with linear equations *(SE p. 125)*	3.1.4 Guided Instruction (10 min.)	Prove angle relationships using linear pairs and vertical angles *(SE p. 130)*
	3.1.2 Exploration (10-15 min.)	Explore complementary angles *(SE p. 126)*	3.1.5 Try It (5 min.)	Solve problems using angle relationships *(SE p. 131)*
	3.1.3 Exploration (10 min.)	Explore supplementary angles *(SE p. 128)*	3.1.6 Wrap-Up (~5 min.)	Use angle relationships to determine values *(SE p. 132)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Complementary Angles • Congruent Complements Theorem • Corollary • Corollary to Congruent Complements Theorem • Supplementary Angles • Congruent Supplements Theorem • Corollary to Congruent Supplements Theorem • Linear Pair • Linear Pair Postulate • Vertical Angles • Vertical Angles Theorem <u>Additional Terms:</u> N/A	• Patty Paper • Compass • Ruler • Yardstick • (Optional) Highlighters or Colored Pens	• Prepare the letters A, B, C, and D for reviewing 3.1.1 Warm-Up, if time permits.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may confuse complementary and supplementary angles.
Students may struggle to understand corollaries and when to use them in proving relationships.</td><td>If not started already, consider having the students start a Proof Resource Booklet with definitions, postulates, and theorems for students to refer to when writing proofs.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Algebra Talk
In 3.1.2 and 3.1.3 Explorations, students are asked to provide details and justifications for angle relationships. Encourage students to seek out others in their class who draw conclusions and/or use reasoning that differs from theirs, and write alternate reasoning in their workbook.</td><td>MA.K12.MTR.5.1: Patterns and Structure
The 3.1.2 and 3.1.3 Explorations and 3.1.4 Guided Instruction provide opportunities for students to work to decompose complex questions into manageable pieces and look for similarities among questions. Students relate previously learned concepts to new concepts. Use these lesson components to develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Notice the 3.1.2 and 3.1.3 Explorations provide auditory and visual entry-points, because students are given the opportunity to listen to other students' conclusions about angles, write down their own summary of their reasoning, and then read through the response to confirm the correct interpretation.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.1.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students choose which one doesn't belong given four linear equations, and provide a justification for their selection.*

3.1.1 Warm-Up: Which One Doesn't Belong?

Four equations are shown.

$$y = 2x$$

$$y = x + 9$$

$$y = -4x + 2$$

$$y = 3x - 2$$

1. Of the four equations, choose one that does not belong.
   Answers will vary.

2. Explain your reasoning for choosing the equation you did.
   Answers will vary.
   $y = 2x$ *It is the only one with only one term.*
   $y = x + 9$ *It is the only one with a coefficient of* 1.
   $y = -4x + 2$ *It is the only one with a negative slope.*
   $y = 3x - 2$ *It is the only one with a negative constant (or* y*-intercept).*

If time permits, consider labeling each equation with a letter (A, B, C, D), and labeling each corner of the room with the same letters. Have students stand in the corner of the room of the equation they chose. Have groups share out why their equation is the one that doesn't belong.

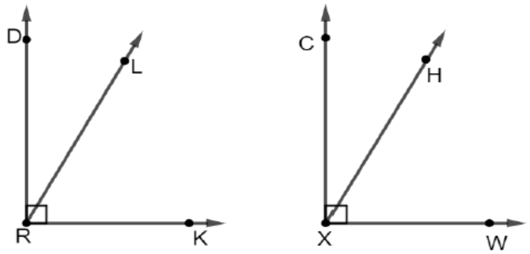
3.1.2 Exploration: Complementary Angles

Component Overview: *Students use the definition of complementary angles to explore other angle relationships. Students use tools as they explore the development of the Congruent Complements Theorem and the Corollary to the Congruent Complements Theorem.*

3.1.2 Exploration: Complementary Angles

Complementary angles are two angles whose measures sum to $90°$.

1. Right angles DRK and CXW are shown. $\overrightarrow{RL}$ is inside $\angle DRK$ and $\overrightarrow{XH}$ is inside $\angle CXW$. $m\angle DRL = 28°$ and $m\angle CXH = 28°$.



a. Use the definition of complementary angles to explain why $\angle LRD$ and $\angle LRK$ are complementary angles.
Given that $\angle DRK$ is a right angle, by the definition of a right angle $m\angle DRK = 90°$.
$\angle LRD$ and $\angle LRK$ are adjacent angles where $m\angle LRD + m\angle LRK = m\angle DRK$.
Therefore, $m\angle LRD + m\angle LRK = 90°$ and $\angle LRD$ and $\angle LRK$ are complementary angles.

b. Determine the measure of $\angle LRK$.
$62°$

c. Discuss with your partner why $m\angle HXW$ must be the same as $m\angle LRK$. Summarize your discussion.
Answers may vary.
$m\angle HXW$ must be the same as $m\angle LRK$ because both $\angle HXW$ and $\angle LRK$ are complementary to a $28°$ angle.

3.1.2 and 3.1.3 Explorations are scaffolded to allow students to engage with them with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered at the conclusion of each component.
Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share before 3.1.4 Guided Instruction.
Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

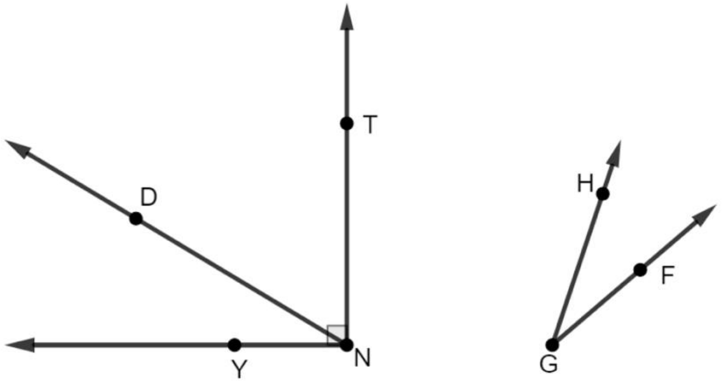
Consider asking the following:

- *What does it mean for a ray to be "inside" an angle?*
- *How did you determine the measure of $\angle LRK$?*

Consider having students share out their reasoning for questions 1a and 1c. Encourage the use of correct vocabulary and the recalling of definitions. For example, "The definition of a right angle says an angle's measure is $90°$; therefore, the two angles that make up $\angle DRK$ must sum to $90°$."

3.1.2 Exploration: Complementary Angles (continued)

2. $\angle YNT$ and $\angle FGH$ are shown. $\overrightarrow{ND}$ is inside $\angle YNT$, such that $m\angle YND = 31°$.



a. What relationship do $\angle YND$ and $\angle DNT$ have?
They are complementary angles.

b. Complete the paragraph proof.

$\angle YNT$ is a(n) [○ acute / ○ right / ○ obtuse] angle. Therefore, by

definition, $m\angle YNT = 90°$. By the angle addition postulate, $m\angle YND + m\angle DNT = m\angle YNT$.

Therefore, by the substitution property of equality, $m\angle YND + m\angle DNT = 90°$.

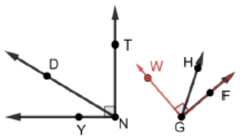
By the definition of [○ supplementary / ○ complementary] angles,

$\angle YND$ and $\angle DNT$ are [○ supplementary / ○ complementary] angles.

While monitoring student work, do not discourage other correct answers for question 2a, such as "they are adjacent" or "they have a sum of 90°."

c. On a piece of patty paper, trace $\angle YND$. Determine which rigid transformations can be used to map $\angle YND$ onto $\angle FGH$.
Answers will vary. Translate $\angle YND$ so point N coincides with point G. Rotate $\angle YND$ counterclockwise about point G until it coincides with $\angle FGH$.

d. Use patty paper or a compass to create $\angle FGW$ so that $m\angle FGW = m\angle YNT$ and $\overrightarrow{GH}$ is inside of $\angle FGW$.



e. Complete the paragraph proof.

Since $m\angle FGW = m\angle YNT$ and $m\angle YNT = 90°$, then $m\angle FGW = 90°$ by the transitive property of equality. By the angle addition postulate, $m\angle FGH + m\angle WGH = m\angle FGW$.

Therefore, by the transitive property of equality, $m\angle FGH + m\angle WGH = 90°$ and are

- supplementary
- complementary

angles by the definition of

- supplementary
- complementary

angles.

Since $m\angle YND + m\angle DNT = 90°$, then by the transitive property of equality $m\angle FGH + m\angle WGH = m\angle YND + m\angle DNT$. Since $m\angle YND = m\angle FGH$, by the substitution property $m\angle FGH + m\angle WGH = m\angle FGH + m\angle DNT$.

Therefore, $m\angle WGH = m\angle DNT$ by the subtraction property of equality.

Encourage students to highlight the angles $\angle YND$ and $\angle FGH$ so that they can clearly distinguish between the angles, so that it's easier to identify the rigid transformations. For students who struggle with tracing, consider providing a copy of $\angle YND$ already drawn on patty paper.

3.1.2 Exploration: Complementary Angles (continued)

Congruent Complements Theorem

Complements of congruent angles are congruent.

3. Use the congruent complements theorem to complete the statements.

   $\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle MCY$.
   $\angle THM$ is congruent to $\angle WBK$, so $\angle LGX$ is congruent to $\angle MCY$.

A ***corollary*** is a true statement that is a simple deduction from a theorem or postulate. Its proof requires only a few simple statements in addition to the proof of the original theorem or postulate.

Corollary to Congruent Complements Theorem

Complements of the same angle are congruent.

4. Use the corollary to congruent complements theorem to complete the statement.

   $\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle LGX$, so $\angle THM$ is congruent to $\angle WBK$.

For questions 3 and 4, consider prompting students to draw the angles and what their relationships look like before filling in the sentences. Then, have them highlight the congruent angles in the same color to better see the angle relationships.

3.1.3 Exploration: Supplementary Angles

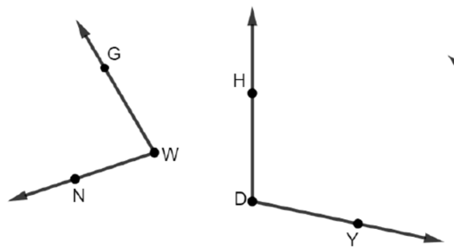
Component Overview: *Students explore supplementary angles as they develop understanding of the Congruent Supplements Theorem and its corollary.*

3.1.3 Exploration: Supplementary Angles

Supplementary angles are two angles whose measures sum is exactly $180°$.

For a visual entry point, have students copy $\angle HDY$ on patty paper and use the copy to show the angle is supplementary to both $\angle GWN$ and to $\angle KFX$.

1. $\angle GWN$, $\angle HDY$, and $\angle KFX$ are shown. $\angle GWN$ is supplementary to $\angle HDY$. $\angle KFX$ is supplementary to $\angle HDY$.



a. Determine what relationship, if any, $\angle GWN$ and $\angle KFX$ have.
$\angle GWN$ and $\angle KFX$ are congruent.

b. Discuss with your partner what can be concluded about $\angle GWN$ and $\angle KFX$. Summarize your discussion.
Answers will vary. If both $\angle GWN$ and $\angle KFX$ are supplementary to the same angle, then we can conclude:

- *they must measure less than* $180°$.
- *they must have the same measure.*

c. Ask two classmates their conclusion about $\angle GWN$ and $\angle KFX$. Record their conclusion and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

3.1.3 Exploration: Supplementary Angles (continued)

Conclusion	My Summary of Their Reason	Initials
Answers will vary.		

d. Discuss your classmates' conclusions with your partner.

Congruent Supplements Theorem

Supplements of congruent angles are congruent.

2. Use the congruent supplements theorem to complete the statements.
$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle MCY$. $\angle LXG$ is congruent to $\angle BKW$, so $\angle HTM$ is congruent to $\angle MCY$.

Corollary to Congruent Supplements Theorem

Supplements of the same angle are congruent.

3. Use the corollary to congruent supplements theorem to complete the statement.
$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle LXG$, so $\angle HTM$ is congruent to $\angle BKW$.

For an auditory entry-point and for students who have difficulties recording verbal responses, consider allowing the student with the conclusion to share their conclusion verbally, and then have the other student repeat back to the first student what they heard. Then, the students can work together to edit their responses to question 1b, if needed.

When students are discussing their classmates' conclusions with their partners, encourage pairs to notice any common conclusions and how the summary of their reasons is similar or different.

Both the Corollary to Congruent Complements Theorem and the Corollary to the Congruent Supplements Theorem are seen as extensions of the original Theorem. When using these as reasons in a proof, the original theorem can be used for both.

3.1.4 Guided Instruction: Angles

***Component Overview**: Students use the Congruent Supplements Theorem to complete a proof determining linear pairs are congruent. The result of this proof is meant to lead a student to notice the Vertical Angles Theorem as well.*

3.1.4 Guided Instruction: Angles

Two adjacent angles that form a straight angle are called a linear pair.

A ***linear pair*** is two adjacent angles formed by two intersecting lines.

Linear Pair Postulate

If two angles form a linear pair, then they are supplementary.

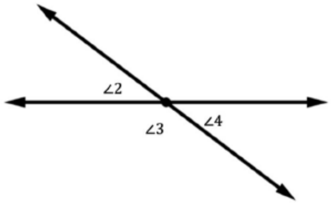
When introducing this component, consider having an image projected of two intersecting lines with all four angles numbered 1-4. Have students name all the adjacent angles first, then name all of the linear pairs. Use a yardstick or large sheet of paper to highlight each angle pair while hiding the other angles for an additional visual entry-point.

1. Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank	
Definition of Linear Pair	Linear Pair Postulate
Supplements of the same angle are congruent.	Complements of the same angle are congruent.

3.1.4 Guided Instruction: Angles (continued)

Given:
$\angle 2$ and $\angle 3$ form a linear pair.
$\angle 3$ and $\angle 4$ form a linear pair.
Prove: $\angle 2 \cong \angle 4$



Statement	Reason
1. $\angle 2$ and $\angle 3$ form a linear pair	1. Given
2. $\angle 3$ and $\angle 4$ form a linear pair	2. Given
3. $\angle 2$ and $\angle 3$ are supplementary angles	*3. If two angles form a linear pair, they are supplementary.*
4. $\angle 3$ and $\angle 4$ are supplementary angles	*4. If two angles form a linear pair, they are supplementary.*
5. $\angle 2 \cong \angle 4$	*5. Supplements of the same angle are congruent*

Vertical angles are the opposite angles formed when two lines intersect.

2. $\angle 2$ and $\angle 4$ are also known as *vertical* angles.

Vertical Angles Theorem

If two lines intersect, then vertical angles are congruent.

Before moving to the two-column proof, have students mark the given information on the image. Then, have students discuss the relationship between $\angle 2$ and $\angle 4$ and explain their thinking to each other.

Ask students to explain why the definition of a linear pair and the Linear Pair Postulate are not the same thing, as well as the definition of vertical angles and the Vertical Angles Theorem. Sample responses could include that the definition describes the position of the angles, while the theorem describes the relationship between the angles.

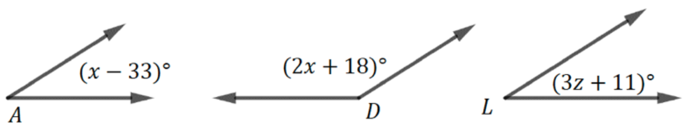
3.1.5 Try It: Solving Problems Using Angle Relationships

Component Overview: *Students apply the angle relationships explored in the lesson to find the measures of unknown variables and angles. Consider allowing students to work with a partner.*

3.1.5 Try It: Solving Problems Using Angle Relationships

1. Three angles are shown with the given information.

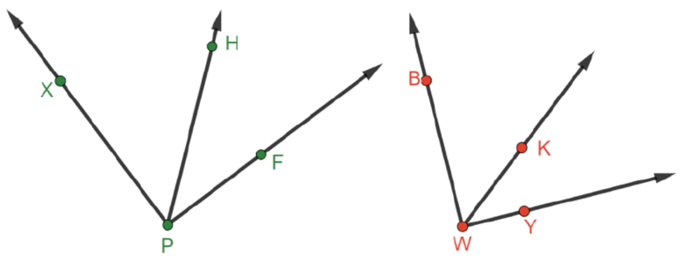
$\angle A$ is supplementary to $\angle D$
$\angle L$ is supplementary to $\angle D$



What is the value of z?
$(x-33)+(2x+18)=180$
$x=65$
$(x-33)=(3z+11)$
$65-33=3z+11$

$z=7$

2. Right angles XPF and $\angle BWY$ are shown. $\overrightarrow{PH}$ is inside $\angle XPF$, such that $m\angle HPF = 39°$. $m\angle HPF = m\angle KWY$.



What is $m\angle BWK$?
$51°$

Allow informal observation up to this point in the lesson to dictate how "guided" this component should be for your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

Consider asking the following if students struggle getting started:

- *What does it mean to be supplementary?*
- *Are there any angles that are congruent? How do you know?*
- *Can you solve an equation with two different variables?*
- *Which expressions are written in terms of the same variable? What is the relationship between the angles that these expressions represent?*

For a challenge, have students write a proof in any form (two-column, paragraph, or Flow Chart) for question 2.

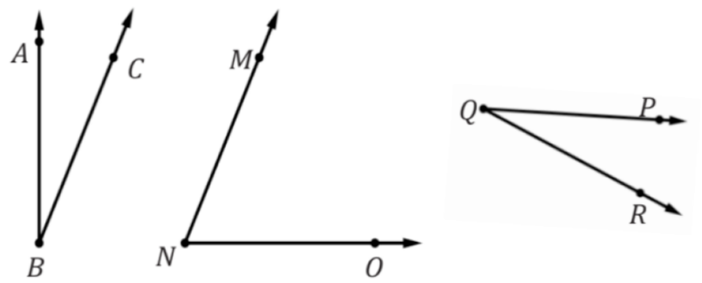
3.1.6 Wrap-Up: Cool Down

Component Overview: *Students find the value of a variable and the measure of an angle given information about three angles.*

3.1.6 Wrap-Up: Cool Down

1. Three angles are shown with the given information.

$\angle ABC$ is complementary to $\angle MNO$
$\angle MNO$ is complementary to $\angle PQR$
$m\angle ABC = (5x - 3)°$
$m\angle MNO = 2(x + 22)°$.



a. Find the value of x.
7

b. What is $m\angle PQR$?
32°

If using this as a formative assessment, make note of where students make mistakes for developing remediation steps.

- *If students set the two given expressions equal to each other, then they may misunderstand the definition of complementary angles.*
- *If students set the equation up correctly but solve incorrectly, they may need support on solving equations. Check the On-Ramp area for resources to remediate solving equations.*
- *If students solve part a correctly, but answer part b as 58°, they may misunderstand the Congruent Complements Theorem.*

3.2 Vertical and Corresponding Angles

Lesson Introduction

Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.	Learning Targets	• I can use vertical angles to solve problems. • I can use corresponding angles formed by parallel lines and a transversal to solve problems.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How are corresponding angles formed by parallel lines and a transversal related? • How can you use vertical angles to solve problems?
Lesson Narrative	This lesson begins with mapping a line onto another parallel line through a rigid transformation to explore the angle relationship formed by a transversal crossing those lines. Through this rigid transformation, students determine that corresponding angles are congruent, leading to the Corresponding Angles Theorem. The corresponding angle relationships formed from two parallel lines and a transversal are used, along with vertical angles, to determine the measures of other angles formed by two parallel lines, intersected by a transversal.

Component Summary				
	3.2.1 Warm-Up (~5 min.)	Determine a value in a sequence from given shapes *(SE p. 133)*	3.2.3 Guided Practice (10-15 min.)	Learn to use the Corresponding Angles Theorem to determine unknown values and angles *(SE p. 135)*
	3.2.2 Exploration (10-15 min.)	Explore corresponding angles formed by parallel lines and a transversal *(SE p. 133)*	3.2.4 Your Turn! (10 min.)	Use relationships between vertical angles and corresponding angles to solve problems *(SE p. 136)*
	3.2.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 138)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Transversal • Corresponding Angles • Corresponding Angles Theorem Additional Terms: • Complementary • Supplementary • Congruent	• (Optional) Patty Paper	• (Optional) Manipulatives or technology to demonstrate angle relationships formed between parallel lines and a transversal.

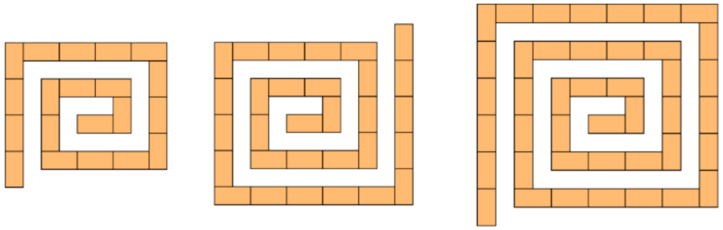
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may confuse corresponding and vertical angles because they both are congruent to the original angle.
Students may not set the sum of linear pair angles equal to $180°$.</td><td>While this lesson does not require the use of patty paper, students may need it to see the translations of angles to show that they are corresponding.
Consider using objects or technology to demonstrate how the corresponding angles are congruent to each other, regardless of the angle by which the transversal crosses the parallel lines. This can be demonstrated using three wooden dowels, pencils, two lines drawn on the board and a yardstick, etc.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Multiple components of this lesson create opportunities for students to discuss their ideas and thinking with their partners. 3.2.2 Exploration and in 3.2.3 Guided Practice are scaffolded so that students are given the opportunity to think about the problems, and then share and discuss them with their partner.</td><td>MA.K12.MTR.5.1: Patterns and Structure
3.2.3 Guided Practice question 1b has students using the structure of the problem to connect mathematical concepts. Students will focus on the details of the problem and relate the previously learned concepts of linear pairs and supplementary angles to describe how to determine the measure of an angle from given information. Provide opportunities for students to create a plan to solve the problem.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider opportunities in this lesson where students can first identify the angle relationship before finding a missing value. This allows students to structure their thinking when approaching these types of problems, so that with future problems, this type of support is not needed.
Create a kinesthetic entry-point in 3.2.2 Exploration by allowing students to use patty paper to explore transformations involving angles formed by parallel lines and a transversal. Allow students to copy one angle and determine which other angles are congruent by moving the copied angle on top of the other angles through a series of rigid motions.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.2.1 Warm-Up: Shape Sequence

Component Overview: *Students are asked to look at a pattern and determine the number of rectangles that will follow in the sequence.*

3.2.1 Warm-Up: Shape Sequence

A sequence of three shapes created using rectangles is shown.



1. How many rectangles will have to be added for the next step in the sequence?
   14 *rectangles*

2. How many rectangles will be in Figure 11?
   210 *rectangles*

For students who finish early, consider asking any of the following questions as extensions:

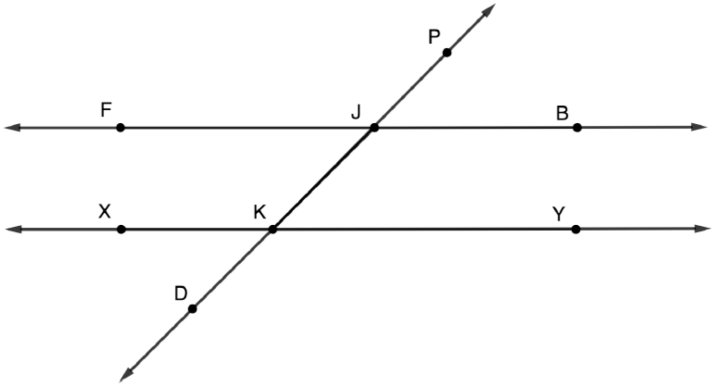
- *How many rectangles are in figure* 20*?*
- *Can you write an expression that can be used to find the number of rectangles in any figure,* n*?*

3.2.2 Exploration: Parallel Lines

Component Overview: *Students explore relationships when parallel lines are crossed by a transversal, with a focus on corresponding and vertical angles.*

3.2.2 Exploration: Parallel Lines

$\overleftrightarrow{FB}$, $\overleftrightarrow{XY}$, and $\overleftrightarrow{DP}$ are shown, where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{FB}$ at point J and where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{XY}$ at point K.



1. Discuss with your partner what rigid transformations can be used to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K.

2. How can you use a translation to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K?
   Translate $\overleftrightarrow{FB}$ down a number of units and to the left a number of units.

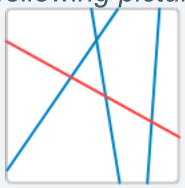
In the diagram, $\overleftrightarrow{PD}$ is also known as a transversal, and $\overleftrightarrow{FB}$ and $\overleftrightarrow{XY}$ are parallel lines denoted as $\overleftrightarrow{FB} \| \overleftrightarrow{XY}$.

A ***transversal*** is a line that intersects two or more lines in the same plane at different points.

3.2.2 Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before the 3.2.3 Guided Practice. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

If students are having difficulties noticing a rigid transformation, providing patty paper can help them better visualize the transformation.

To extend the students' thinking, consider drawing the following picture on the board:



Ask students to discuss with their partner which line(s) would be considered transversal using the definition given.

3.2.2 Exploration: Parallel Lines (continued)

3. Use the translation of $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K to name the new angles in the chart.

Pre-image	Image
$\angle PJF$	$\angle PKX$
$\angle PJB$	$\angle JKY$
$\angle DJB$	$\angle DKY$
$\angle FJD$	$\angle XKD$

The angles listed in the table above are corresponding angles.

Corresponding angles are angles that are in the same position on two lines in relation to a transversal.

4. Complete the statement.

Given that corresponding angles of parallel lines can be mapped to each other using rigid transformations, it can be concluded that the angles are

- complementary.
- congruent.
- supplementary.

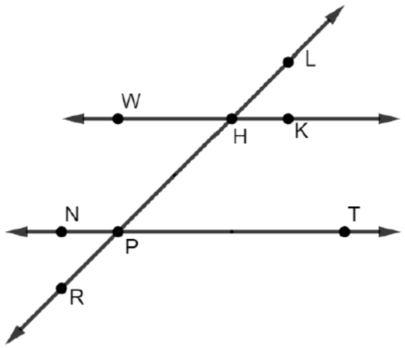
Consider providing students with patty paper for this component to add a kinesthetic entry-point.

Students may use different points to name the angles in question 3. Have students find a classmate that named the angle differently. Have them discuss why they could both have different answers but still be correct.

Based on the statement from question 4, what can be said about $\angle PJF$ and $\angle JKX$?

3.2.2 Exploration: Parallel Lines (continued)

5. In the figure shown, $\overleftrightarrow{WK} \parallel \overleftrightarrow{NT}$ where $\overleftrightarrow{PH}$ is a transversal.



a. For each angle, name a corresponding angle and a vertical angle.

	$\angle TPR$	$\angle WHP$	$\angle KHP$	$\angle NPR$
Corresponding	$\angle KHP$	$\angle NPR$	$\angle TPR$	$\angle WHP$
Vertical	$\angle HPN$	$\angle LHK$	$\angle LHW$	$\angle HPT$

b. Discuss your answers with your partner and make changes, if necessary.

Before students start on question 5a, consider utilizing a Think-Pair-Share asking students to discuss what they know about corresponding and vertical angles. Listen for academic language such as "intersect" and "opposite angles." Consider having students list similarities and differences between vertical angles and corresponding angles if time allows.

Similarities:

- *Formed by intersecting lines*
- *Congruent angles*

Differences:

- *Vertical angles are formed by two intersecting lines, and corresponding angles are formed by two parallel lines and an intersecting transversal.*

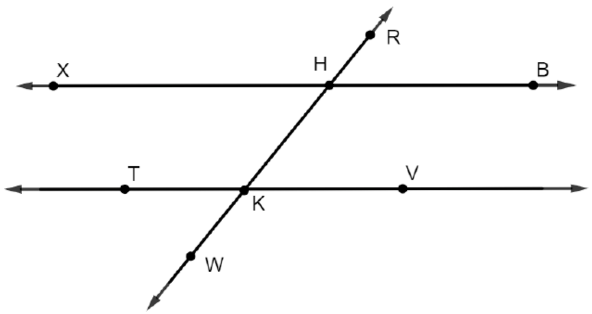
3.2.3 Guided Practice: Angles in Parallel Lines

Component Overview: *Students apply the relationships between corresponding angles explored in the previous component to determine the angle measures of angles formed by two parallel lines and a transversal.*

3.2.3 Guided Practice: Angles in Parallel Lines

Corresponding Angles Theorem
If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown such that $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ is a transversal.



a. Given $m\angle RHB = 51°$, determine the measurement of each angle.

$m\angle HKV = 51°$

$m\angle TKW = 51°$

$m\angle XHK = 51°$

b. Describe how to use $m\angle RHB = 51°$ to find $m\angle XHR$.
Because $\angle XHR$ and $\angle RHB$ are adjacent angles formed by two intersecting lines, then they are a linear pair. The linear pair postulate states that two angles that form a linear pair are supplementary. Supplementary angles sum to $180°$ by definition so $m\angle XHR + m\angle RHB = 180°$. Therefore $m\angle XHR + 51° = 180°$ and $m\angle XHR = 129°$ by the subtraction property of equality.

For a visual entry-point before students start identifying the angle measure, consider prompting them to highlight all angles congruent to $\angle RHB$. Doing this first will have students focusing on the angle relationships before considering numerical values.

As students share out their answers, listen for and reinforce the use of academic vocabulary already introduced (adjacent angles, intersecting lines, linear pair, supplementary). Once a student shares a correct response, have another student repeat what they said using the correct vocabulary. This move builds toward the justification needed in writing proofs.

3.2.3 Guided Practice: Angles in Parallel Lines (continued)

c. Explain why knowing $m\angle XHR$ allows all of the angles to be determined.
Once you know $m\angle XHR$, the rest of the angles can be determined using the Vertical Angles Theorem and the Corresponding Angles Theorem.

2. $\overleftrightarrow{MN}$, $\overleftrightarrow{DK}$, and $\overleftrightarrow{PW}$ lie on a plane such that $\overleftrightarrow{MN} \parallel \overleftrightarrow{DK}$ and $\overleftrightarrow{PW}$ is a transversal. $\overleftrightarrow{PW}$ intersects $\overleftrightarrow{MN}$ at point Y and $\overleftrightarrow{DK}$ at point F. $m\angle PYN = (4x + 36)°$, $m\angle NYF = (11y)°$, and $m\angle KFW = 44°$.

a. Draw a diagram that represents the given information. Answers will vary.

Diagram 1

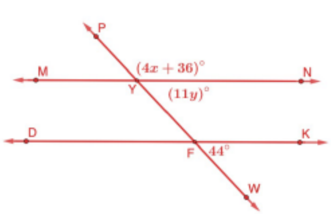
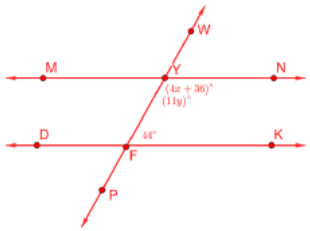


Diagram 2



b. Determine the value of y.
Answers will vary.
Diagram 1 $y = 4$
Diagram 2 $y = 12\frac{4}{11}$

c. Determine the value of x.
$x = 25$

d. What is $m\angle PYM$?
$44°$

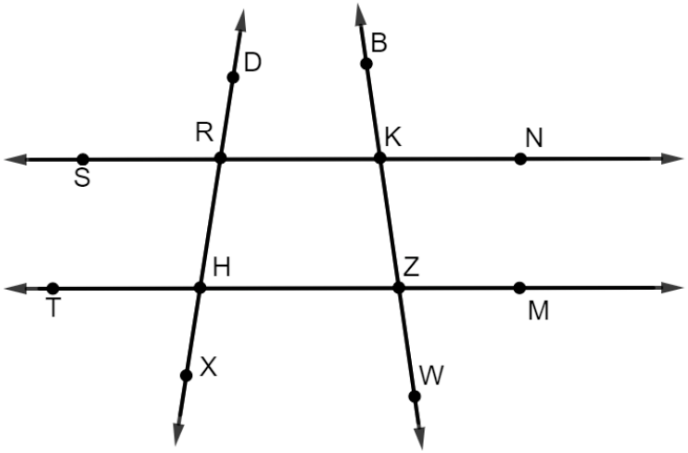
Monitor students as they draw and label the parallel lines and transversal with the appropriate information for question 2a. Consider prompting students to compare their drawings with a partner and resolve any differences prior to proceeding to question 2b - 2d.

3.2.4 Your Turn!

Component Overview: *Students use relationships between vertical angles and corresponding angles to solve questions. Questions include analyzing the reasoning of others.*

3.2.4 Your Turn!

1. $\overleftrightarrow{SN}$, $\overleftrightarrow{TM}$, $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ lie on a plane, such that $\overleftrightarrow{SN} \parallel \overleftrightarrow{TM}$ and $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ are transversals. $m\angle BKN = 99°$ and $m\angle KRH = 99°$.



a. What is $m\angle KZM$?
$99°$

b. What is $m\angle ZHX$?
$99°$

c. What is $m\angle DRS$?
$99°$

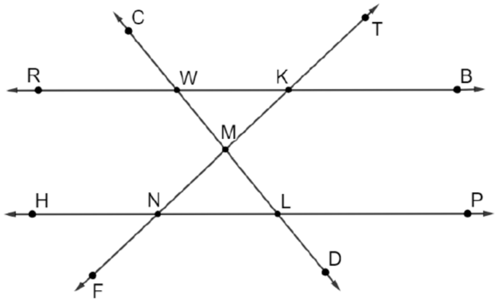
For a visual entry-point, consider having students first mark all angles on the figure congruent to $\angle BKN$ and then congruent to $\angle KRH$ in a different color, before determining the values of the angle measures.

Consider having them share their markings with a partner, explaining how they know the angles are congruent.

For an extension, have students write a justification of their answers. For example, $m\angle KZM$ is $99°$ because it is a corresponding angle to $\angle BKN$, formed by transversal $\overleftrightarrow{BW}$ intersecting parallel lines $\overleftrightarrow{SN}$ and $\overleftrightarrow{TM}$.

3.2.4 Your Turn! (continued)

2. $\overleftrightarrow{CD}$, $\overleftrightarrow{TF}$, $\overleftrightarrow{RB}$, and $\overleftrightarrow{HP}$ are shown where $\overleftrightarrow{RB} \parallel \overleftrightarrow{HP}$ and $m\angle CWR = 51°$ and $m\angle TKB = 43°$.



Jasmyn's and Drew's justifications of the measure of $\angle CLH$ is shown.

Jasmyn's Answer	Drew's Answer
$m\angle CLH = 51°$ because $\angle CLH$ and $\angle CWR$ are corresponding angles.	$m\angle CLH = 43°$ because $\angle TKB$ and $\angle CLH$ are corresponding angles.

a. Explain which student you agree with, including your reasoning.
Answers will vary. I agree with Jasmyn because $\angle CLH$ and $\angle CWR$ are corresponding angles. Drew is incorrect because $\angle TKB$ and $\angle CLH$ do not share a transversal, therefore they cannot be corresponding angles.

b. Find the measure of the following angles.

$m\angle KNL = \underline{43°}$

$m\angle HNF = \underline{43°}$

$m\angle MLN = \underline{51°}$

Consider having students edit Drew's answer to make it correct.

3.2.5 Wrap-Up: Cool Down

Component Overview: *Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

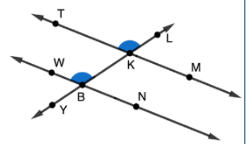
3.2.5 Wrap-Up: Cool Down

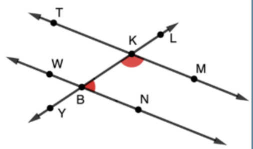
1. Draw an example of vertical angles.
   Answers will vary.

2. Draw an example of corresponding angles.
   Answers will vary.

3.3 Congruent Angles in Parallel Lines

<table>
<tr><th colspan="6">Lesson 3 Introduction</th></tr>
<tr><td>Benchmark Focus</td><td colspan="2">MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.</td><td>Learning Target</td><td colspan="2">• I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On Working Toward</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• What are the names of the angle pairs formed by parallel lines intersected by a transversal?
• What is the relationship between angle pairs formed by two parallel lines intersected by a transversal?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">This lesson begins with students using patty paper to determine the relationships between angles formed by two parallel lines and a transversal. Students are introduced to alternate interior and alternate exterior angles. They identify angle pairs and complete a formal proof of the Alternate Exterior Angle Theorem. Practice centers around determining the measures of other angles when given the measure of one angle formed by parallel lines and transversals.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>3.3.1 Warm-Up (~5 min.)</td><td>Career profile video using math to make guitars (SE p. 139)</td><td>3.3.3 Try It (10 min.)</td><td colspan="2">Determine angle measures when parallel lines are intersected by a transversal (SE p. 142)</td></tr>
<tr><td>3.3.2 Guided Instruction (15-20 min.)</td><td>Explore alternate interior and alternate exterior angle relationships (SE p. 139)</td><td>3.3.4 Your Turn! (5 min.)</td><td colspan="2">Practice determining angle measures when parallel lines are intersected by a transversal (SE p. 143)</td></tr>
<tr><td>3.3.5 Wrap-Up (~5 min.)</td><td colspan="4">Ticket Out the Door identifying angle relationships and using to find angle measure (SE p. 143)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td>Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms:</u>
• Alternate Exterior Angles Theorem
• Alternate Interior Angles Theorem
<u>Additional Terms:</u> N/A</td><td>• Patty Paper
• (Optional) Highlighters</td><td colspan="2">• 3.3.1 Warm-Up includes the students watching a career profile video.
• Consider providing a word bank for students to choose from for the proof in 3.3.2 Guided Instruction.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may confuse the name of alternate interior and alternate exterior angles, or the relationship between the angle pairs.
• Students may assume the same angle relationships exist between angles formed by two different transversals.</td><td>• Congruence of the angle pairs only exist when the lines intersected by the transversal are parallel. Consider providing examples of non-parallel lines intersected by a transversal to show that alternate interior angles, alternate exterior angles, and corresponding angles do not have congruence in this case.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
3.3.3 Try It question 1 offers an opportunity for students to share what they notice and wonder about a given figure before completing a table. Allowing students to make sense of the image prior to answering the questions provides an entry-point into the question.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Question 2 of 3.3.3 Try It involves parallel lines with two intersecting transversals. Offering a challenging task fosters perseverance as students analyze and make sense of the problem. Students should be encouraged by recognizing their efforts when presented a challenge. Use this as an opportunity to develop students' ability to analyze and problem-solve based on the relationships they have learned.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The use of patty paper in the 3.3.2 Guided Instruction creates visual and kinesthetic entry-points as students make sense of the angle relationships formed by two parallel lines intersected by a transversal. Some students struggle with accurately tracing images and may benefit from already having two angles drawn to match the two angle measures shown in the figure.



</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.3.1 Warm-Up: Using Math in Making Guitars

Component Overview: *Students watch a career video of how mathematics is used in building guitars.*

3.3.1 Warm-Up: Using Math in Making Guitars

Patrick Evans is a production manager at Maton Guitars in Australia. Country music star Keith Urban uses Maton guitars. The precision necessary in making each guitar requires mathematics, or "maths," as Australians call it.

1. What was one thing you found interesting about the process of making a guitar?
   Answers will vary.

Consider having the career profile video "Maths: Make your career count - Guitar Maker" (4:17 minutes) preloaded to avoid potential technology challenges.

What other connections can you make between music and mathematics?

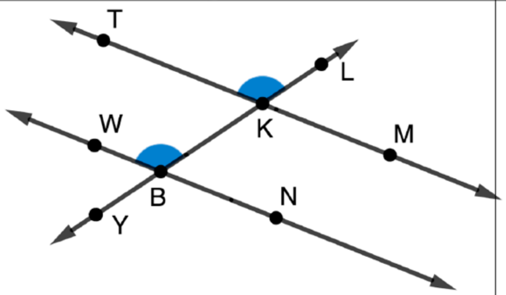
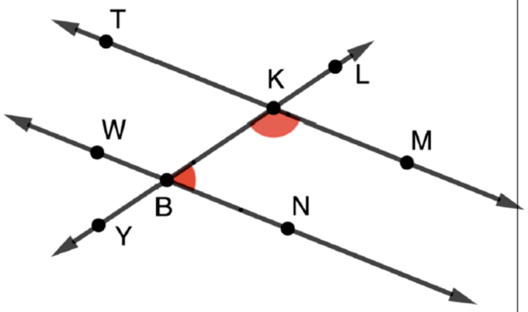
3.3.2 Guided Instruction: Angle Relationships

Component Overview: *Students investigate the relationship between angles formed by two parallel lines intersected by a transversal before being introduced to the Alternate Exterior Angles Theorem and the Alternate Interior Angles Theorem. After identifying different angle pairs, students complete a two-column proof of the Alternate Exterior Angles Theorem.*

3.3.2 Guided Instruction: Angle Relationships

1. In each diagram, $\overleftrightarrow{TM} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{YL}$ is a transversal.

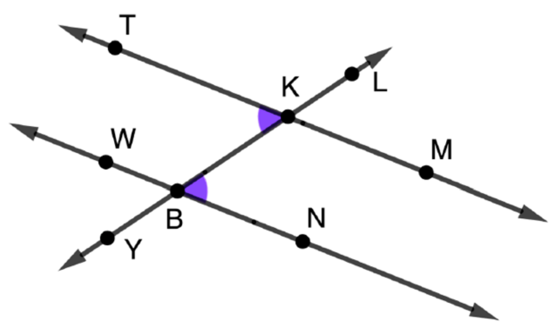
   a. Use patty paper to determine if the angles marked using color are congruent.

Diagram	Congruent?
A 	○ Yes ○ No
B 	○ Yes ○ No

For students who struggle with tracing images, consider providing copies of the two differently sized angles shown in the images for students to use in the investigation.

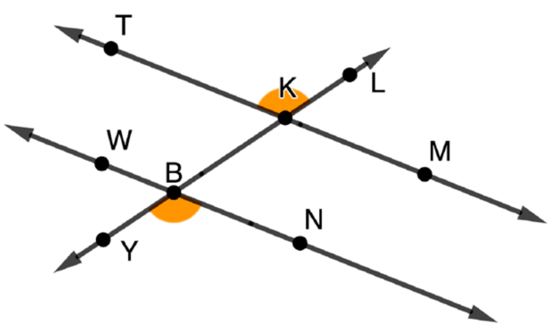
3.3.2 Guided Instruction: Angle Relationships (continued)

C



○ Yes
○ No

D



○ Yes
○ No

Before moving to the following theorems, engage students in a discussion about the placement of these four angle pairs in relation to the parallel lines and the transversal.

- *Are the angles on the interior or the exterior of the parallel lines?*
- *Are the angles on the same side of the transversal or are they on opposite sides of the transversal (alternate)?*

Having students make observations about the locations of the angles in relation to each other will help students understand the names of the angle pairs.

Which pair of angles from question 1a shows corresponding angles?

Alternate Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.

b. Which diagram shows a pair of angles that are alternate exterior angles?

- Diagram A
- Diagram B
- Diagram C
- Diagram D

What is another pair of alternate exterior angles in diagram D?

Alternate Interior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.

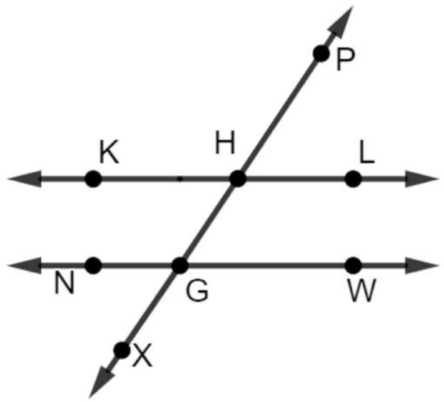
c. Which diagram shows a pair of angles that are alternate interior angles?

- Diagram A
- Diagram B
- Diagram C
- Diagram D

What is another pair of alternate interior angles in diagram C?

3.3.2 Guided Instruction: Angle Relationships (continued)

2. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$, where $\overleftrightarrow{XP}$ is a transversal.

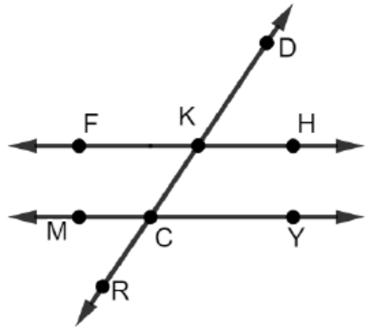


Work with your partner to name each pair of angles as corresponding, alternate exterior, alternate interior angles, or none of these.

Angles	Type
$\angle WGX$ and $\angle KHP$	○ Corresponding angles ○ Alternate exterior angles ○ Alternate interior angles ○ None of the above
$\angle LHG$ and $\angle WGX$	○ Corresponding angles ○ Alternate exterior angles ○ Alternate interior angles ○ None of the above
$\angle NGX$ and $\angle KHP$	○ Corresponding angles ○ Alternate exterior angles ○ Alternate interior angles ○ None of the above
$\angle KHG$ and $\angle WGH$	○ Corresponding angles ○ Alternate exterior angles ○ Alternate interior angles ○ None of the above

For students that are having difficulties seeing and differentiating between the angle relationships, consider providing highlighters or colored markers so students can use one color for each angle pair. This will allow students to see the locations of the angles more clearly.

3. Given: $\overleftrightarrow{FH} \parallel \overleftrightarrow{MY}$
Prove: $\angle DKH \cong \angle RCM$



Fill in the table to complete the proof.

Statement	Reason
1. $\overleftrightarrow{FH} \parallel \overleftrightarrow{MY}$	1. Given
2. $\angle DKH \cong \angle FKC$	2. Vertical angles are congruent.
3. $\angle CKF \cong \angle RCM$	3. *If two parallel lines are intersected by a transversal, then corresponding angles are congruent.*
4. $\angle DKH \cong \angle RCM$	4. Transitive Property of Congruence

The intention of question 3 is to guide students through the completion of the two-column proof. Consider modeling the reasoning used throughout the process as a think-aloud, allowing opportunities for students to ask questions. If you prefer to release this proof to students first, consider the differentiation ideas below.

If desired, consider providing the opportunity for students to first try it with a partner on their own before reviewing whole-group. In this case, consider providing a word bank for students with the statements and reasons.
You could create the support as you see fit. For students who need a lot of support, you can provide only the statements and reasons that will be used in the proof. Then, as they go through the proof, they can cross out the ones they used.

For students who do not need as much of the support, you can still provide statements and reasons, but also include statements and reasons that won't be used. Some examples to include that will not be used in the proof could be:

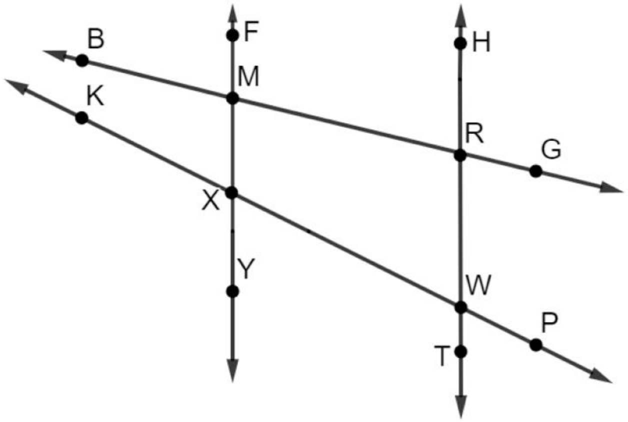
- $\angle MCR \cong \angle KCY$
- $\angle DKF \cong \angle YCR$
- *Linear pairs are supplementary*
- *Corresponding angles are supplementary*

3.3.3 Try It: Angles and Parallel Lines

Component Overview: *Students will work to identify the measures of angles created by the intersection of a transversal and two parallel lines and provide a justification of their reasoning.*

3.3.3 Try It: Angles and Parallel Lines

1. $\overleftrightarrow{FY} \parallel \overleftrightarrow{HT}$ where $\overleftrightarrow{BG}$ and $\overleftrightarrow{KP}$ are transversals and $m\angle BMF = 76°$ and $m\angle KXY = 117°$.



Complete the table.

Angle	Angle Measure
$\angle MRW$	$104°$
$\angle PWT$	$63°$
$\angle YXW$	$63°$
$\angle RMX$	$76°$
$\angle PWR$	$117°$

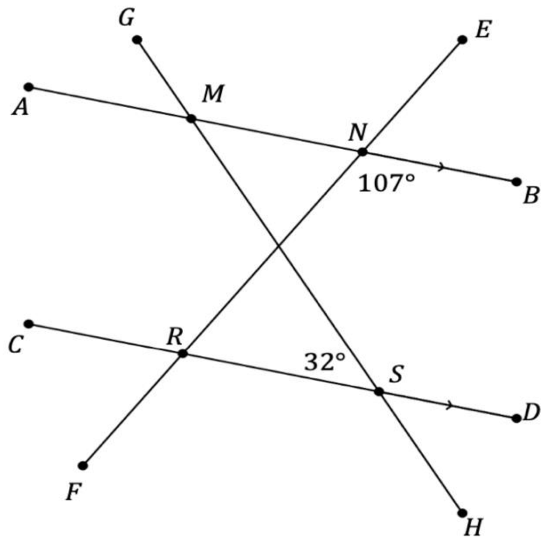
Notice and Wonder
Consider implementing this strategy for students to share what they notice and wonder before completing the table. Allowing students to make sense of the image provides an entry point into the problem.
Things that students may notice include:

- *Only two of the lines are parallel.*
- *There are two different transversals.*
- *The angle measures formed by one transversal are not congruent to the angles formed by the other transversal.*
- *There is a quadrilateral in the middle of the image, but it is not a parallelogram (this is an important observation that will be explored later in the course).*

For students who are having difficulties finding congruent angles, have them cover up one of the transversals to find special angle pairs formed by the two parallel lines and one of the transversals.

3.3.3 Try It: Angles and Parallel Lines (continued)

2. $\overline{AB} \parallel \overline{CD}$ where $\overline{GH}$ and $\overline{EF}$ are transversals and $m\angle BNR = 107°$ and $m\angle RSM = 32°$.



Complete the table.

Angle Measure	Postulate or Theorem Used
$m\angle NRC = 107°$	*Alternate Interior Angles Theorem*
$m\angle BMH = 32°$	*Alternate Interior Angles Theorem*
$m\angle DRF = 107°$	*Corresponding Angles Theorem or Vertical Angles Theorem*
$m\angle GMA = 32°$	*Corresponding Angles Theorem or Vertical Angles Theorem*

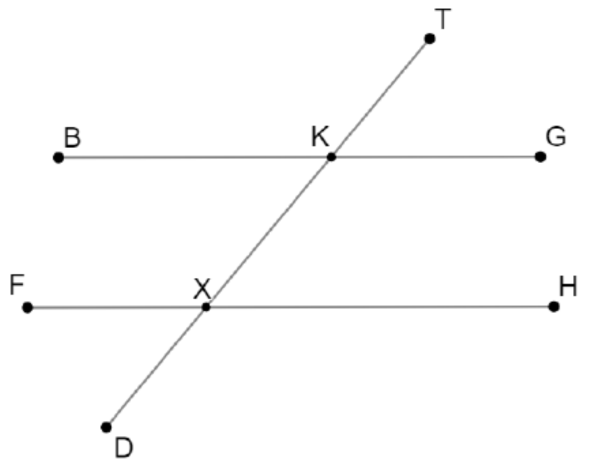
For a challenge, have students label the point where $\overline{MS}$ and $\overline{NR}$ meet as point Q. Then, have them determine if it is possible to find $m\angle MQN$? Have students support their answers with angle relationships.

3.3.4 Your Turn!

Component Overview: *Students will work to identify the angle measures and the angle relationship of angles created by the intersection of a transversal and two parallel lines.*

3.3.4 Your Turn!

1. $\overline{BG} \parallel \overline{FH}$ where $\overline{TD}$ is a transversal and $m\angle HXD = 134°$.



Find the measure of each angle. Then, state the angle relationships used to determine the value.

Answers will vary.

Angle Measure	Angle Relationships
$m\angle GKX = 134°$	*Corresponding Angles*
$m\angle FXK = 134°$	*Vertical Angles*
$m\angle BKT = 134°$	*Alternate Exterior Angles*

Consider using this time to work with small groups or individuals that are still having difficulties identifying these relationships and using them to solve problems.

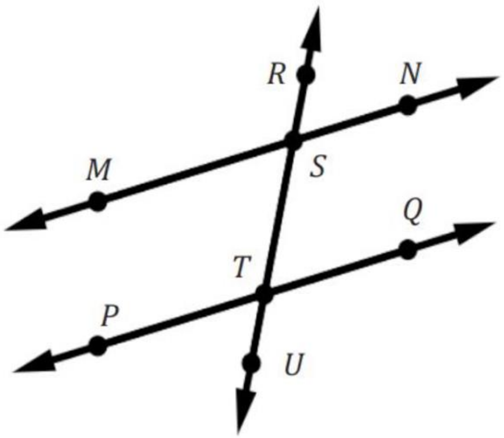
After we find the measure of these three angles, are we able to find the measure of the rest of the angles? Explain and support your answer.

3.3.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students will demonstrate their understanding of angle pair relationships created by two parallel lines intersected by a transversal.*

3.3.5 Wrap-Up: Ticket Out the Door

1. $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$ and $m\angle PTS = 100°$.



   a. In relation to the parallel lines, what angle relationship do $\angle TSN$ and $\angle PTS$ have?
   They are alternate interior angles. They are congruent angles.

   b. What is $m\angle TSN$?
   $100°$

Proficiency on this Wrap-Up includes:

- *Identifying the angle relationship between two angles*
- *Using the angle relationship to determine the measure of an unknown angle*

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.

3.4 Consecutive Angles

Lesson 4 Introduction

Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.	Learning Targets	• I can determine the relationships of consecutive exterior and interior angles. • I can solve problems that involve angles formed by parallel lines intersected by a transversal.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• What relationship do consecutive interior angles have? • What relationship do consecutive exterior angles have? • How can you use angle relationships to find unknown values and unknown angle measures?
Lesson Narrative	In this lesson, the angle relationships of consecutive interior angles and consecutive exterior angles created by two parallel lines intersected by a transversal are established. Practice includes using these relationships along with the angle relationships introduced in the prior lessons to solve for unknown values and unknown angle measures.

Component Summary	3.4.1 Warm-Up (~5 min.)	Connect numbers with their factors through area *(SE p. 144)*	3.4.3 Guided Practice (10 min.)	Determine values of unknowns and angle measures using angle relationships *(SE p. 148)*
	3.4.2 Collaboration (25-30 min.)	Prove the relationships of Consecutive Exterior Angles and Consecutive Interior Angles Theorems *(SE p. 145)*	3.4.4 Wrap-Up (~5 min.)	Reflect on confidence with angle relationships explored in the unit *(SE p. 149)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Consecutive Interior Angles Theorem • Consecutive Exterior Angles Theorem <u>Additional Terms:</u> N/A	• (Optional) Patty Paper	• Consider providing a word bank for students to choose from when completing the proofs in 3.4.2 Collaboration.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may confuse consecutive interior and exterior angles with corresponding angles or alternate interior or exterior angles. • Students may relate an angle measure to angles created with the wrong transversal.	• Students may get the various angle definitions mixed up, either because they sound similar, or because they have similar locations. Consider providing a Word Wall with the definitions and pictures to support students as they learn this academic vocabulary.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time In this lesson, students are provided with an opportunity to refine their verbal and written responses with questions found in the 3.4.2 Collaboration. Students will discuss with partners and others in the class to compare and revise their conjectures.	**MA.K12.MTR.1.1: Participate in Effortful Learning** 3.4.2 Collaboration provides an opportunity for students to analyze an image to discover the relationships of consecutive interior and exterior angles. Encourage students to help and support each other when attempting a new method or approach. Students also build their perseverance in 3.4.3 Guided Practice by completing questions involving multiple transversals and angle measures represented by algebraic expressions.
Differentiate Instruction	Consider providing patty paper in this lesson so that students can more visually see the congruence of angles, and that consecutive interior and exterior angles sum to $180°$. Word Walls or word banks can also be used for scaffolded support of students when completing proofs and describing angle relationships in 3.4.2 Collaboration and 3.4.3 Guided Practice. 3.4.2 Collaboration allows students to discuss and correct their answers through auditory and visual entry points when discovering the relationships of consecutive interior and consecutive exterior angles.	

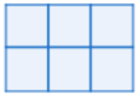
Lesson Notes:

3.4.1 Warm-Up: Squares in Rectangles

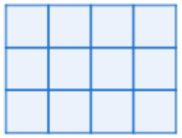
Component Overview: *Students make the connection between the dimensions of a rectangle and its area and how the possible dimensions are the factors of the area of the rectangles.*

3.4.1 Warm-Up: Squares in Rectangles

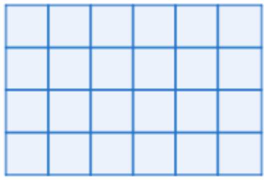
Three rectangles and their descriptions are shown.



A 2×3 rectangle contains 6 unit squares.



A 3×4 rectangle contains 12 unit squares.



A 4×6 rectangle contains 24 unit squares.

1. What are the dimensions of a rectangle that contains exactly 100 unit squares?
   A 10×10 square.

2. List any other size rectangles that contain exactly 100 unit squares.
   Answers will vary.
   $1 \times 100;\ 2 \times 50;\ 4 \times 25;\ 5 \times 20$

\

For students who finish early, consider extending these questions by having students consider the following statement and defend it as always, sometimes, or never true:

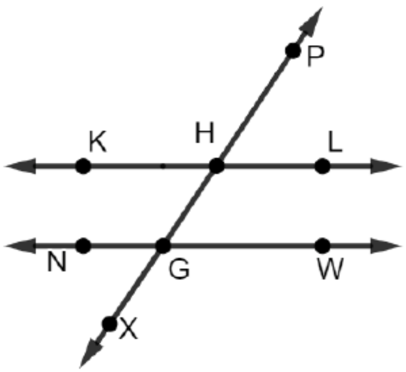
> *"The larger the number of squares a rectangle has, the more options there are for the dimensions of the rectangle."*

3.4.2 Collaboration: Consecutive Interior and Exterior Angles

Component Overview: *Students use an image of parallel lines intersected by a transversal to determine the congruence of pairs of angles and their angle pair name based on their location in the image. Students then write a conjecture about the relationships of consecutive interior angles and consecutive exterior angles, verifying their conjectures by completing a two-column proof before applying the relationships to solve questions.*

3.4.2 Collaboration: Consecutive Interior and Exterior Angles

1. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{XP}$ is a transversal.



a. Work with your partner to determine if each pair of angles could be congruent by using corresponding, alternate exterior, alternate interior, or vertical angles.

Angles	Congruent?
$\angle NGH$ and $\angle KHG$	*NOT congruent*
$\angle PHL$ and $\angle WGX$	*NOT congruent*

b. Work with your partner to determine which pair of angles in the table could be described as consecutive interior angles.
$\angle NGH$ and $\angle KHG$ are consecutive interior angles.

c. Name another pair of consecutive interior angles.
$\angle WGH$ and $\angle LHG$ are also consecutive interior angles.

3.4.2 Collaboration is scaffolded to allow students to engage with it with their partner or small group, rather than being guided through by the teacher.
Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.

Consider having a Word Wall or poster with the various angle relationships that have been taught in the last few lessons, including a visual of each. This will allow students to have a reference point when working through this component with their partner.

As students start questions 1b and 1c, suggest having students breakdown the term "consecutive interior angles" to determine a description of the angle pairs prior to naming the angles.

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

d. Work with your partner to write a conjecture about the relationship that is true for consecutive interior angles.
Conjecture: Consecutive interior angles are supplementary.

e. Ask another classmate for their conjecture about consecutive interior angles. Record their conjecture and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
Answers will vary.		

f. Compare the conjecture from your classmate with the conjecture you and your partner wrote. If necessary, make any adjustments to your conjecture.
Answers will vary.

g. $\angle PHL$ and $\angle WGX$ are consecutive exterior angles. Name another pair of consecutive exterior angles.
$\angle XGN$ and $\angle PHK$ are consecutive exterior angles.

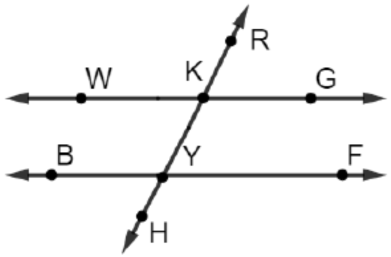
h. Work with your partner to write a true conjecture about the relationship between consecutive exterior angles.
Conjecture: Consecutive exterior angles are supplementary.

What are the differences and similarities between consecutive interior and consecutive exterior angles?

2. A partially completed proof is shown.

   a. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} || \overleftrightarrow{BF}$
Prove: $\angle WKY$ and $\angle BYK$ are supplementary.



Statement	Reason
1. $\overleftrightarrow{WG} \| \overleftrightarrow{BF}$	1. Given
2. $\angle WKY \cong \angle BYH$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle BYH$ and $\angle BYK$ are a linear pair	3. Definition of a linear pair
4. *$\angle BYH$ and $\angle BYK$ are supplementary*	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle BYH + m\angle BYK = 180°$	5. Definition of supplementary angles
6. $m\angle WKY = m\angle BYH$	*6. Definition of congruence*
7. $m\angle WKY + m\angle BYK = 180°$	7. Substitution
8. $\angle WKY$ and $\angle BYK$ are supplementary.	8. Definition of supplementary angles

   b. Check your proof with another classmate. If necessary, make any adjustments.

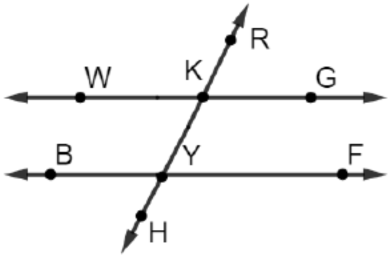
Consider providing a word bank for students who are struggling to fill out the proof.
Below is a possible word bank to use. Angles can be used more than once.

- BYH
- WKY
- $\angle BYH$
- $\angle WKY$
- $m\angle BYH$
- $m\angle WKY$
- *Definition of congruence*
- *Definition of supplementary angles*
- *Definition of complementary angles*
- *$\angle BYH$ and $\angle BYK$ are supplementary*
- *$\angle BYH$ and $\angle FYH$ are supplementary*

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

3. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} || \overleftrightarrow{BF}$
Prove: $\angle RKG$ and $\angle FYH$ are supplementary.



Statement	Reason
1. $\overleftrightarrow{WG} \| \overleftrightarrow{BF}$	1. Given
2. $\angle RKG \cong \angle KYF$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle KYF$ and $\angle FYH$ are a linear pair	3. Definition of a linear pair
4. $\angle KYF$ *and* $\angle FYH$ *are supplementary*	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle KYF + m\angle FYH = 180°$	5. Definition of supplementary angles
6. $m\angle RKG = m\angle KYF$	6. *Definition of congruence*
7. $m\angle RKG + m\angle FYH = 180°$	7. Substitution
8. $\angle RKG$ and $\angle FYH$ are supplementary.	8. Definition of supplementary angles

Consider providing a word bank for students who are struggling to fill out the proof.
Below is a possible word bank to use. Angles can be used more than once.

- KYF
- RKG
- $\angle KYF$
- $\angle RKG$
- $m\angle KYF$
- $m\angle RKG$
- $m\angle KYF + m\angle KYB = 180°$
- $m\angle KYF + m\angle FYH = 180°$
- $m\angle RKG \cong m\angle BYH$
- $m\angle RKG \cong m\angle KYF$
- *Definition of congruence*
- *Definition of supplementary angles*
- *Definition of complementary angles*
- $\angle BYH$ *and* $\angle KYB$ *are supplementary*
- $\angle KYF$ *and* $\angle FYH$ *are supplementary*

If students are creating a proof resource booklet or guide, consider having them add common statements that "typically" follow each other in a proof. For example, the following statements typically, but not always, follow each other in a proof:

____ and ____ are linear pairs	*definition of linear pair*
____ and ____ are supplementary	*linear pair postulate*
$\_\_\_\_ + \_\_\_\_ = 180°$	*definition of supplementary angles*

3.4.2 Collaboration: Consecutive Interior and Exterior Angles (continued)

4. Mistakes can be an opportunity to learn.

   a. What mistakes did you make in the first proof that helped you in the second proof?

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

   b. If you did not make a mistake in the first proof, describe how the proofs were similar.
   Answers will vary.

Consecutive Interior Angles Theorem
If two parallel lines are intersected by a transversal, then interior angles on the same side of the transversal are supplementary.

Consecutive Exterior Angles Theorem
If two parallel lines are intersected by a transversal, then exterior angles on the same side of the transversal are supplementary.

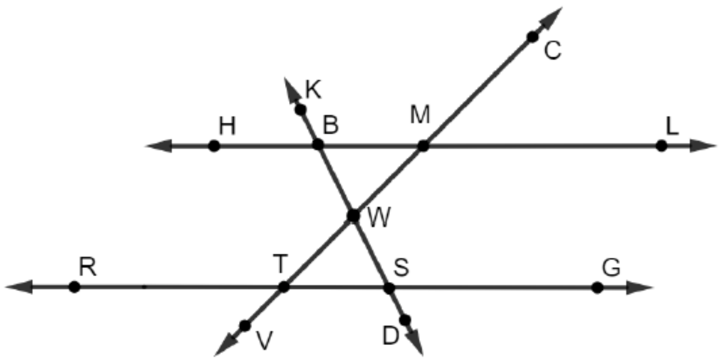
Consider having students compare their mistakes with their partner and discuss with each other how to learn from them to prevent making them in the future.

3.4.3 Guided Practice: Angles in Parallel Lines

Component Overview: *Students apply relationships of angles formed by two parallel lines intersected by a transversal to solve for unknown values and angle measures.*

3.4.3 Guided Practice: Angles in Parallel Lines

1. $\overleftrightarrow{HL} \parallel \overleftrightarrow{RG}$ where $\overleftrightarrow{CV}$ and $\overleftrightarrow{KD}$ are transversals with $m\angle KBM = 117°$ and $m\angle VTS = 134°$.



Determine the measurement of each angle.

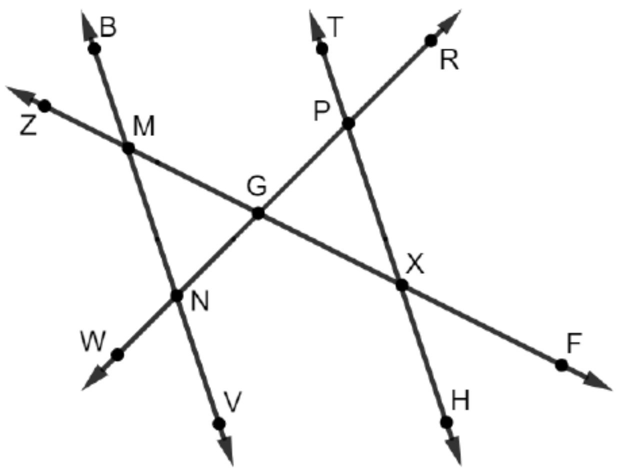
$m\angle DSG = 63°$ $\quad$ $m\angle LMC = 46°$

$m\angle RTV = 46°$ $\quad$ $m\angle BWM = 71°$

Allow informal observation throughout the first two components to dictate how "guided" 3.4.3 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

Consider having students either write or verbalize a justification for how they found each angle measurement. This justification would prevent random guessing on the answer.
An example of a justification could be:
$m\angle DSG = 63°$ *because it is a consecutive exterior angle, and supplementary to* $\angle KBM$ *formed by parallel lines* $\overline{HL}$ *and* $\overline{RG}$*, intersected by transversal* $\overline{KD}$*.*
Having the students name the transversal when two transversals are shown will help to clear misconceptions students have about angle relationships.

2. In the figure shown, $\overleftrightarrow{BV} \parallel \overleftrightarrow{TH}$, $m\angle GXH = (5(2x-1))°$, $m\angle GMN = (4x-11)°$, $m\angle XPG = (3y-8)°$, and $m\angle VNG = (7y-52)°$.



   a. Find the value of x.
   $(4x-11)+(5(2x-1))=180$
   $x=14$

   b. Find the value of y.
   $(7y-52)+(3y-8)=180$
   $y=24$

Before having students start on question 2, consider having them name the angle relationships of the given angles along with the transversal.
This support will allow students who are still confused on how to apply the angle relationships, and provides a way for them to map out their thinking.

3.4.4 Wrap-Up: Reflection

Component Overview: *Students reflect on their confidence and understanding of the angle relationships explored in Lessons 1 – 4.*

3.4.4 Wrap-Up: Reflection

1. Rate your confidence in solving problems using these angle pairs.

Answers will vary.

I can solve number problems involving….	I am an expert and could help others.	I can hold my own.	I am getting there.	I would like more help or practice.
Congruent Supplements				
Congruent Complements				
Vertical Angles				
Consecutive Interior Angles				
Consecutive Exterior Angles				
Alternate Interior Angles				
Alternate Exterior Angles				
Corresponding Angles				

What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of these angle relationships thus far in the unit and how best to address potential gaps in understanding in upcoming lessons.

3.5 Converse of the Alternate Interior Angles Theorem

Lesson Introduction

<table>
<tr><td>Benchmarks</td><td>MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all,” and “not” statements. Find the converse, inverse, and contrapositive of a statement.</td><td>Learning Targets</td><td colspan="2">I can write and use the converse of the postulate or theorem.
I can use postulates and theorems about parallel lines to determine if lines are parallel.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td>Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td>MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How do you write the converse of a statement?
What angle relationships can be used to determine if two lines are parallel?
What angle relationships cannot be used to determine if two lines are parallel?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">The lesson connects the postulates and theorems learned in the previous lessons to their converses. Students will determine and justify whether lines are parallel given the relationship of angle pairs formed by two lines intersected by a transversal. Students will explore which information is sufficient to prove lines are parallel and which is not.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>3.5.1 Warm-Up (~5 min.)</td><td>Notice and Wonder about lines in a video game (SE p. 150)</td><td>3.5.3 Activity (15-20 min.)</td><td>Gallery Walk exploring the converses of parallel line postulates and theorems (SE p. 151)</td></tr>
<tr><td>3.5.2 Exploration (5-8 min.)</td><td>Conjecture if given angle relationships are sufficient to determine lines are parallel (SE p. 150)</td><td>3.5.4 Guided Practice (10-12 min.)</td><td>Determine if lines are parallel using angle relationships (SE p. 152)</td></tr>
<tr><td>3.5.5 Wrap-Up (~5 min.)</td><td colspan="3">Students summarize their learning (SE p. 153)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
Congruent
Converse
Parallel Lines</td><td>(Optional) Chart Paper for Gallery Walk</td><td colspan="2">Consider having a Word Wall or Anchor Chart in the classroom that includes images of the various angle relationships created by parallel lines intersected by a transversal.</td></tr>
</table>

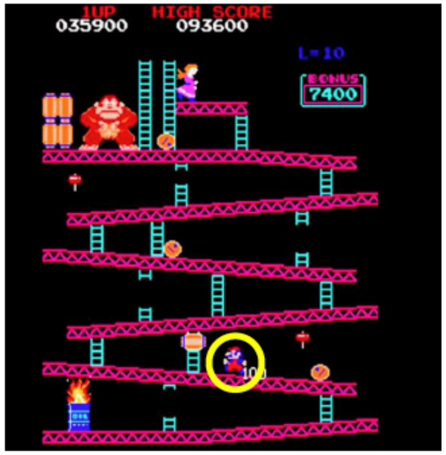
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may incorrectly say lines are parallel based on linear pairs being supplementary and vertical angles being congruent.</td><td>When students struggle justifying lines are parallel when given multiple transversals, consider having students make a list of the angles formed by each transversal prior to answering any questions. This move will help students see the angle pairs before determining the relationships.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
In 3.5.4 Guided Practice, have students employ the Take Turns routine when sharing their response in the table. Using this routine limits the likelihood that one student takes over while the other student does not participate. If students disagree with their partner’s answer, they should provide a rationale for their disagreement and resolve it through discussion.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
Throughout the lesson, students are asked to communicate mathematical ideas, vocabulary involving angle relationships, and their methods for justifying if two lines are parallel. During the Gallery Walk in the 3.5.3 Activity, students will also analyze the statements of others to determine if it is correct and/or what corrections need to be made.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The 3.5.3 Activity provides both auditory and visual entry points. Students discuss the converse of each postulate and theorem with a partner and provide a visual example.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.5.1 Warm-Up: Notice and Wonder

Component Overview: *Students consider a video game screen and write down what they notice and wonder.*

3.5.1 Warm-Up: Notice and Wonder

The game *Donkey Kong* was released for sale in the U.S. in 1981 by Nintendo, a Japanese playing card and toy company turned video game developer. Mario, circled in yellow, is shown in the image running and climbing up the platforms and ladders.



1. What do you notice about the different platforms Mario is running on?
   Answers will vary. They represent line segments that are not parallel to each other.

2. What do you wonder?
   Answers will vary. Can Mario climb any of the ladders?

3. How can you determine if any of the platforms or ladders are parallel?
   Answers will vary. I can determine if the ladders are parallel by measuring the angles formed by the platforms and the ladders to determine the angle measure relationships.

Consider having students compare what they noticed in question 1 with their partner and write down one thing that their partner noticed that they did not.

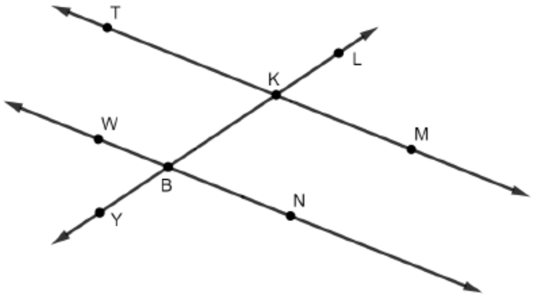
If time allows, consider asking students to share their responses to question 3 with a partner or with the whole group as a transition to the next component.

3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

Component Overview: *Students will work with a partner to conjecture if there is enough information to determine if two lines are parallel from various pieces of given information.*

3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

1. $\overleftrightarrow{TM}$, $\overleftrightarrow{WN}$, and $\overleftrightarrow{YL}$ are shown. $\overleftrightarrow{YL}$ intersects $\overleftrightarrow{TM}$ and $\overleftrightarrow{WN}$.



Work with your partner to conjecture if the given information in each row of the table is sufficient to show that the lines are parallel.

Angles	Type of Angles	Sufficient?
$\angle TKL \cong \angle WBK$	*Corresponding angles*	○ Yes ○ No
$\angle MKB \cong \angle WBK$	*Alternate interior angles*	○ Yes ○ No
$\angle MKB \cong \angle TKL$	*Vertical angles*	○ Yes ○ No
$\angle TKL \cong \angle YBN$	*Alternate exterior angles*	○ Yes ○ No
$\angle WBK$ and $\angle NBK$ are supplementary.	*Linear pair*	○ Yes ○ No
$\angle TKL$ and $\angle WBY$ are supplementary.	*Consecutive exterior angles*	○ Yes ○ No

For the angles that do not have sufficient evidence to show that the lines are parallel, have students explain their reasoning. Listen for evidence that includes the angles being formed by one line and the transversal, and not two lines and a transversal.

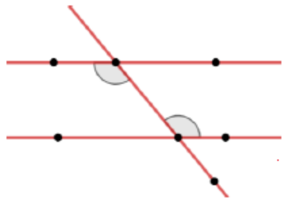
At the conclusion of 3.5.2 Exploration, consider guiding students through a discussion of their answers, if time allows. This can make for a rich mathematical discussion, as well as address any potential misconceptions.

3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

Component Overview: *In this component, students write converses of postulates or theorems and draw pictorial representations.*

3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

1. Work with your partner or group to write the converse of the postulate or theorem your group is assigned. Then, draw a picture to represent the relationship described.

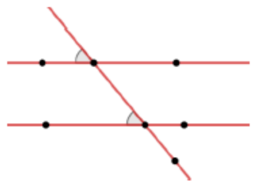
Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*
Picture	
	

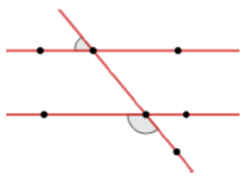
Consider structuring 3.5.3 Activity as a Gallery Walk. Assign each group or student pair a postulate or theorem and have them write the converse and draw a pictorial representation on large sheets of paper. Once done, have the students participate in a Gallery Walk where each group presents their representation. Students record the representations in their workbooks.

If students do not recall how to write a converse statement, consider prompting them to revisit Lesson 4 from Unit 2 in their workbook.

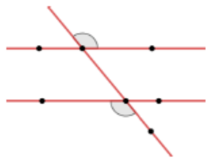
When two lines are intersected by a transversal, if the angle pair relationships of parallel lines and a transversal occur, then the lines are parallel. Each angle pair relationship formed by parallel lines intersected by a transversal postulate and theorem can be written as converses.

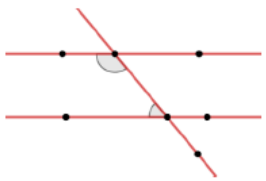
3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems (continued)

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then corresponding angles are congruent.	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.*
Picture	
	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.	*If two lines are intersected by a transversal so that consecutive exterior angles are supplementary, then the lines are parallel.*
Picture	
	

3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems (continued)

<table>
<tr><th>Postulate or Theorem</th><th>Converse</th></tr>
<tr><td>If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.</td><td>If two line are intersected by a transversal so that alternate exterior angles are congruent, then the lines are parallel.</td></tr>
<tr><th colspan="2">Picture</th></tr>
<tr><td colspan="2"></td></tr>
</table>

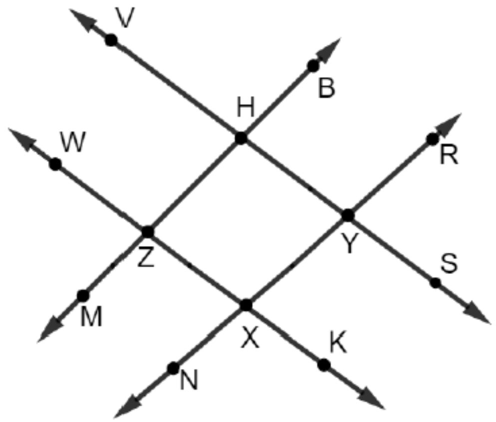
<table>
<tr><th>Postulate or Theorem</th><th>Converse</th></tr>
<tr><td>If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.</td><td>If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.</td></tr>
<tr><th colspan="2">Picture</th></tr>
<tr><td colspan="2"></td></tr>
</table>

3.5.4 Guided Practice: Determining if Lines Are Parallel

Component Overview: *Students use the converse of the angle pair relationship theorems and postulates to determine which lines are parallel, including providing justification.*

3.5.4 Guided Practice: Determining if Lines are Parallel

1. $\overleftrightarrow{VS}$, $\overleftrightarrow{WK}$, $\overleftrightarrow{MB}$, and $\overleftrightarrow{NR}$ are shown where $m\angle VHB = 98°$, $m\angle RYS = 78°$, $m\angle NXK = 102°$, and $m\angle WZM = 82°$.

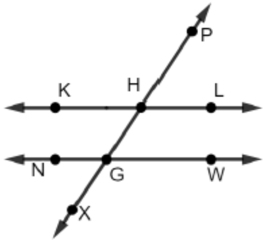


Which pair(s) of lines are parallel? Justify your answer.
Answers will vary. $\overleftrightarrow{VS}$ and $\overleftrightarrow{WK}$ are parallel because $\angle VHB$ and $\angle WZM$ are supplementary consecutive exterior angles formed by the two lines and transversal $\overleftrightarrow{MB}$.

Use observational data from 3.5.2 Exploration and 3.5.3 Activity to determine how much guidance is needed to bring together key ideas in 3.5.4 Guided Practice. If necessary, consider modeling question 1 to address common misconceptions shown in prior components.

For students who finish quickly, consider having them explain why lines $\overleftrightarrow{BM}$ and $\overleftrightarrow{RN}$ are not parallel? Students should provide at least one angle relationship that supports their answer.

2. $\overleftrightarrow{KL}$, $\overleftrightarrow{NW}$, and $\overleftrightarrow{PX}$ are shown.



Determine if the given condition could be used to justify that $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$?	Justification
$\angle PHL \cong \angle WGX$	○ Yes ○ No	*Congruent consecutive exterior angles are not a justification for proving parallel lines.*
$m\angle KHP + m\angle NGX = 180°$	○ Yes ○ No	*Consecutive exterior angles are supplementary.*
$\angle KHG \cong \angle WGH$	○ Yes ○ No	*Alternate interior angles are congruent.*
$\angle NGX$ and $\angle WGX$ are supplementary.	○ Yes ○ No	*$\angle NGX$ and $\angle WGX$ are a linear pair, which is not a justification for proving parallel lines.*
$\angle LHG \cong \angle KHP$	○ Yes ○ No	*$\angle LHG$ and $\angle KHP$ are vertical angles, which is not a justification for proving parallel lines.*

Take Turns
For question 2, consider having students employ the Take Turns routine when sharing their response in the table. Using this routine limits the likelihood that one student takes over while the other student does not participate. If students disagree with their partner's answer, they should provide a rationale for their disagreement and resolve it with a discussion.

3.5.5 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

3.5.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.
   Answers will vary.

3.6 Using Angle Relationships, Postulates and Theorems

Lesson Introduction

Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.	Learning Target	• I can solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.
Benchmark Coherence	Building On	Working Toward	
	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A	
Essential Questions	• How can you use the angle relationships, postulates, and theorems to solve problems?		
Lesson Narrative	This lesson focuses on providing students practice in applying what they have learned about angle relationships to solve problems. The lesson features a Card Sort activity where students collaborate to match images with the name of angle pairs identified. Students will also match an answer card by using the angle pair relationships to create an equation and solve for x. Students then practice applying these relationships to solve problems independently.		

Component Summary				
	3.6.1 Warm-Up (~5 min.)	Would You Rather with two ramps *(SE p. 154)*	3.6.3 Your Turn! (15 min.)	Practice using angle pair relationships to solve problems *(SE p. 156)*
	3.6.2 Activity (20 min.)	Card Sort using angle pair relationships to solve for variables *(SE p. 155)*	3.6.4 Wrap-Up (~5 min.)	Ticket Out the Door using angle pair relationships to find missing angle measures *(SE p. 157)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Congruent • Supplementary • Parallel Lines • Transversal	• Blackline Master	• For 3.6.2 Activity, teachers will need to make copies and cut out columns and rows of the blackline master to create sets to give to pairs or small groups of students.

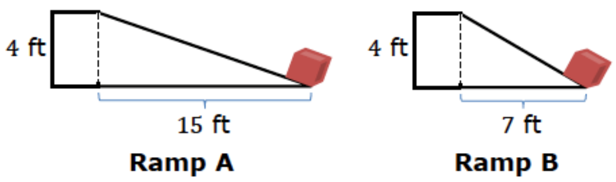
<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may use the incorrect angle pair relationships when creating equations to solve.
Students may make errors related to solving equations.
Students may confuse when they are solving for a variable versus a missing angle.</td><td>Consider using information from the previous lessons to form a small group of students who are struggling working with angle relationships or solving equations. Facilitate these students as they work through 3.6.2 Activity.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematic Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Group Presentations
For the 3.6.2 Activity, consider having a different pair of students present their answers to the class for each matching set. Students should use proper academic language when describing the pairings and when justifying how to create the equations. After a pair finishes, have another student restate why the students made the matches they did.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will use their knowledge of the angle pair relationships formed by two parallel lines and a transversal, linear pairs, and vertical angles to solve problems. Students will need to focus on relevant details within a problem to know how to create the equations and solve them. Support students to develop generalizations based on the similarities found among problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">3.6.2 Activity provides a kinesthetic entry point using a Card Sort. Students will work with a partner or small group to match the name of the angle pair, image of the angle pair, and the value of the variable when solving using the angle pair relationship.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.6.1 Warm-Up: Would You Rather?

Component Overview: *Students choose whether they would rather push a box up of one of two ramps with different dimensions.*

3.6.1 Warm-Up: Would You Rather?

1. Two ramps set up to move boxes up to a 4 foot (ft) high platform are shown.



   a. Would you rather push a 50 pound (lb) box up Ramp A or Ramp B?
   Answers will vary.
   I would rather push my box up Ramp A because it is less steep.
   I would rather push my box up Ramp B because the ramp is shorter.

   b. Justify your choice.
   Answers will vary.
   The slope of Ramp A is $\frac{4}{15}$ or approximately 0.27 making it less steep than Ramp B which has a slope of $\frac{4}{7}$ or approximately 0.57. I would rather push the box up a less steep incline.
   The length of Ramp B is $\sqrt{65}$, or approximately 8, ft long while Ramp A is approximately 15 ft long. I would rather push the box a shorter distance.

Students may compare steepness or may compare length. Both are appropriate with the correct justification.

Do you know someone who is a carpenter, architect, or engineer? Consider inviting a guest speaker to join your class (virtually or in person). Students can ask questions to the guest speaker. If you have a Career Technical Education (CTE) program, have the instructor and some of their students come to your class to discuss this scenario.

3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

Component Overview: *Students will work with a partner or small group to complete a Card Sort. They will match the name of an image with the name of the angle relationship labeled and then use the information to find the matching value for the variable.*

3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

With your partner, sort the cards to match each image with the type of angle pair represented and the value of x. Record your answers in the table, showing your work in the last column.

Type of Angle Pair	Letter of Image	Work and Answer
Vertical Angles	*A*	$x + 25 = 2x + 5$ $20 = x$
Linear Pair	*B*	$(7x - 46) + (3x + 6) = 180$ $x = 22$
Alternate Interior Angles	*C*	$5x - 34 = 3x + 16$ $x = 25$
Alternate Exterior Angles	*D*	$13x + 9 = 14x + 2$ $7 = x$
Corresponding Angles	*E*	$3x - 33 = 2x + 26$ $x = 59$
Consecutive Exterior Angles	*F*	$(3x + 10) + (5x + 90) = 180$ $x = 10$
Consecutive Interior Angles	*G*	$(x + 15) + (3x + 5) = 180$ $x = 40$

Use observational data from the previous lessons to determine a small group of students who may benefit from additional support during this component. Consider pulling this small group of students to work with them through the activity before they practice on their own in the next component.

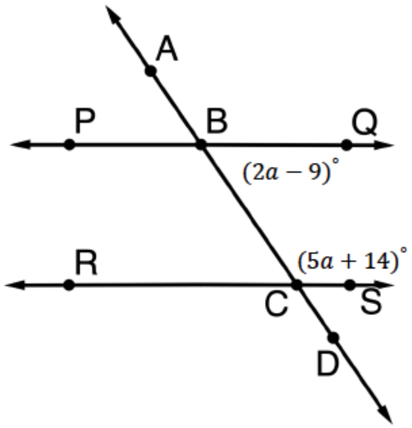
Group Presentations
After completing 3.6.2 Activity for each matching set, consider having a different pair of students present their answers to the class. Students should use proper academic language when describing the pairings and when justifying how to create the equations. After a pair finishes, have another student restate why the students made the matches they did.

3.6.3 Your Turn!

Component Overview: *Students apply what they have learned about various angle pair relationships to solve questions.*

3.6.3 Your Turn!

1. $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$, where $\overleftrightarrow{AD}$ is a transversal.



a. Determine the value of a.
25

b. What is $m\angle ABQ$?
$139°$

c. What is $m\angle BCR$?
$41°$

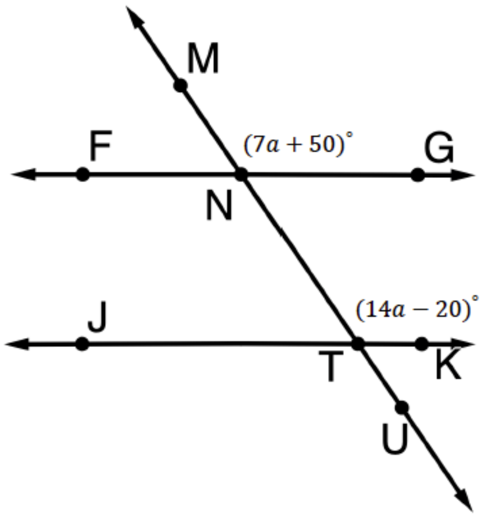
If students are struggling, consider using this time to work with small groups or one-on-one. Consider a peer review process for students to check each other's work as an added layer of support.

While monitoring students, consider asking the following probing questions, as needed:

- *What is the relationship between the measures of these two angle pairs?*
- *What strategy will you use to solve this question?*
- *Why did you pick this strategy?*

3.6.3 Your Turn! (continued)

2. $\overleftrightarrow{FG} \parallel \overleftrightarrow{JK}$ where $\overleftrightarrow{MU}$ is a transversal.



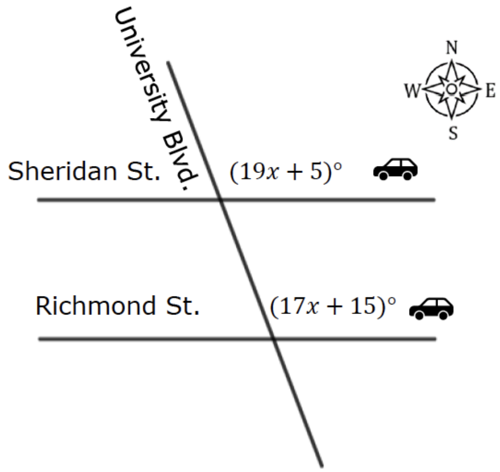
a. Determine the value of a.
10

b. What is $m\angle FNT$?
$120°$

c. What is $m\angle KTU$?
$60°$

For an added challenge, have students justify their answers by organizing their work in a two-column table with each step of their work on the left side, and the explanation on the right.

3. Sheridan Street and Richmond Street are parallel, and both are intersected by University Boulevard.



Two cars are heading west, one on Sheridan St. and one on Richmond St. Both cars turn north onto University Blvd. What is the measure of the angle of their turn onto University Blvd.?

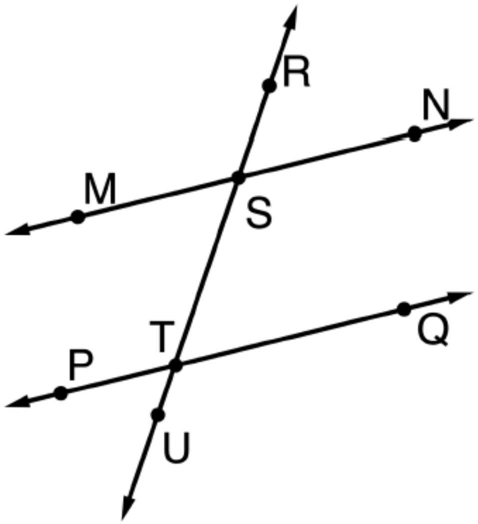
$100°$

3.6.4 Wrap-Up: Ticket Out the Door

Component Overview: *Students apply their knowledge of angle relationships, postulates, and theorems to find missing angle measures.*

3.6.4 Wrap-Up: Ticket Out the Door

1. Given $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$, $m\angle PTS = (14x - 10)°$, and $m\angle NST = (9x + 35)°$, find each angle measure.



a. $m\angle MSR$
$116°$

b. $m\angle STQ$
$64°$

Use the data collected during this activity to develop next steps for student support. Consider the following data points:

- *Did the student label the angles with the correct expression? If not, students may need remediation on naming angles.*
- *Did the student use the correct angle relationship to solve for the variable? If not, students may need remediation on identifying the angle pairs and the theorems associated with them.*
- *Did the student solve the equation correctly? If not, students may need remediation on solving multi-step equations.*
- *Did the student find the angle measure asked for? If not, students may need strategies for reading the question to determine what they are looking for when solving a problem.*

3.7 Solving Problems Using Angle Pair Relationships

Lesson Introduction

Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.	Learning Target	• I can solve mathematical and real-world problems involving angle pair relationships formed by transversals and parallel lines.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How can you use the angle relationships, postulates, and theorems to solve problems?
Lesson Narrative	Students collaborate to solve for variables and missing angles using angle relationships, postulates, and theorems given either mathematical or real-world problems. This lesson is split between partner work and individual work as students apply their knowledge of all angle relationships learned in the unit to solve problems.

Component Summary	3.7.1 Warm-Up (~5 min.)	Match angle pairs based on their relationship in a map *(SE p. 158)*	3.7.3 Your Turn! (15-20 min.)	Practice using angle pair relationships to solve problems *(SE p. 161)*
	3.7.2 Collaboration (15-20 min.)	Use angle relationships to solve for variables and unknown angles *(SE p. 159)*	3.7.4 Wrap-Up (~5 min.)	Reflect on learning targets using a scale *(SE p. 163)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Congruent • Supplementary Angles • Parallel Lines • Transversal	• (Optional) Highlighters or Colored Pencils	N/A

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may use the incorrect angle pair relationships when creating equations to solve. • Students may make errors related to solving equations. • Students may confuse when they are solving for a variable versus a missing angle.	• Consider how you will use the data collected from the 3.6.4 Wrap-Up as well as observational data from Lesson 6 to determine the supports to give to students during this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Consider using this routine in the 3.7.2 Collaboration. The activity is set up to be completed with a partner, but by adding the Think-Pair-Share routine and allowing students to have alone time to work prior to working with their partner, richer discussion that benefit the pair and the class as a whole are more likely to occur.	**MA.K12.MTR.3.1: Mathematical Fluency** The components of this lesson focus on student practice applying what they have learned thus far in Unit 3. Students should select efficient and appropriate methods for solving problems within the given mathematical or real-world context. Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
Differentiate Instruction	3.7.2 Collaboration provides both auditory and visual entry points into discussion. Consider providing roles for students that incorporate multiple entry points by making one student the writer as the other student tells what to write to solve the problem. Students should swap roles for each problem to allow both students the opportunity to participate in each role.	

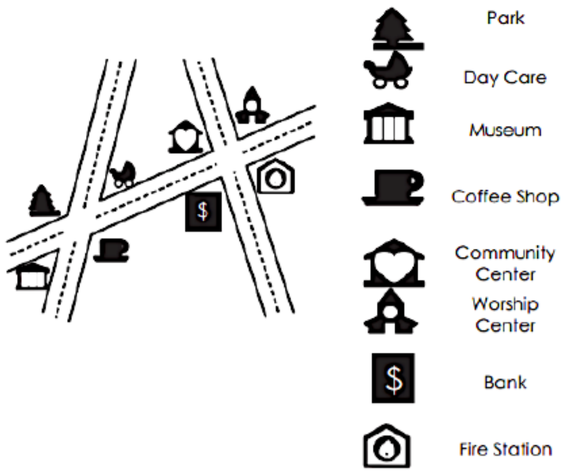
Lesson Notes:

3.7.1 Warm-Up: Angle Pair Relationships

Component Overview: *Students will use a given map to complete this component by naming angle pairs. Students will match different points of interest on the map with the angle relationship the locations form.*

3.7.1 Warm-Up: Angle Pair Relationships

1. The image shows two streets intersected by a third street. Various points of interest are located at the intersections.



Match the listed points of interest with their angle relationship listed in the right column. Write the letter of the angle relationship on the line.

Points of Interest	Angle Relationship
E Museum and Worship Center	A. Alternate Interior Angles
A Bank and Day Care	B. Consecutive Interior Angles
D Park and Coffee Shop	C. Corresponding Angles
F Fire Station and Worship Center	D. Vertical Angles
B Community Center and Day Care	E. Alternate Exterior Angles
C Bank and Museum	F. Linear Pair

If time permits, ask students to confer with a partner about which angle pairs they struggle with identifying. Provide partners the opportunity to share strategies with one another and answer their peers' questions.

3.7.2 Collaboration: Solving Problems Using Angle Relationships

Component Overview: *Students work with a partner to solve mathematical and real-world problems by applying angle relationships, postulates, and theorems in context.*

3.7.2 Collaboration: Solving Problems Using Angle Relationships

Work with your partner to solve each problem.

1. Mako is a steel roller coaster located at Sea World in Orlando. Some of the support beams are shown in the diagram using $\overline{KL}$, $\overline{LR}$, and $\overline{FR}$.

Determine the value of x that shows that $\overline{KL} \parallel \overline{FR}$, if $m\angle KLR = (7x - 5.6)°$ and $m\angle FRL = (3x + 5.6)°$.
$x = 2.8$

When solving multiple-step problems involving angle relationships, consider having students highlight patterns they have seen in the procedures and use them to build their own scaffolding support.

2. Most of the streets in Miami are on a grid. Beacom Blvd. is one of the few streets that does not run north to south or east to west. The diagram shows a map of Beacom Blvd., $\overline{HG}$, between SW 4th Street, $\overline{WT}$, and SW 6th Street, $\overline{BR}$, just a few blocks from Miami Senior High School.

a. Given that $\overline{WT} \parallel \overline{BR}$, determine the value of x if $m\angle HMB = (2x - 7)°$, $m\angle WKG = (4x + 25)°$, and $m\angle KMB = (6x - 29)°$.
$x = 27$

b. Determine the measure of each angle.

$m\angle HMB$
$m\angle HMB = 47°$

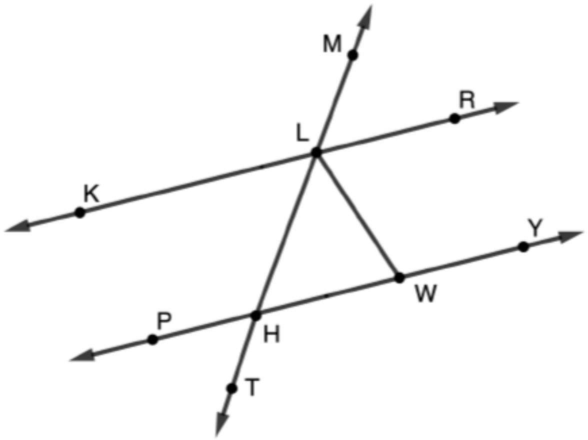
$m\angle HMR$
$m\angle HMR = 133°$

$m\angle TKM$
$m\angle TKM = 133°$

For question 2a, there is more than one way to solve for x based on the given information. While students are working, observe and make note of which students used different methods. Then, after students have completed the question, select students to share their method. Encourage students to provide justification for the steps in their method to provide clarity for those students who are still struggling seeing the angle relationships.

3.7.2 Collaboration: Solving Problems Using Angle Relationships (continued)

3. $\overleftrightarrow{KR}$, $\overleftrightarrow{PY}$, and $\overleftrightarrow{MT}$ are shown with $\overline{LW}$ where $m\angle LWH = (4y - 9\frac{1}{3})°$, $m\angle WLH = (3y + 3)°$, and $m\angle LHW = m\angle WLH$.



a. Determine the value of y that shows that $\overleftrightarrow{PY} \parallel \overleftrightarrow{KR}$.

$y = \frac{55}{3}$

b. Determine the measure of each angle.

$m\angle RLW$

$m\angle RLW = 64°$

$m\angle PHT$

$m\angle PHT = 58°$

$m\angle KLM$

$m\angle KLM = 122°$

Students may not recognize that $\overline{LW}$ is also a transversal of $\overleftrightarrow{KR}$ and $\overleftrightarrow{PY}$. If they miss this relationship, they may struggle with this problem. Listen and watch as students work with their partner to determine whether to provide additional scaffolding support.

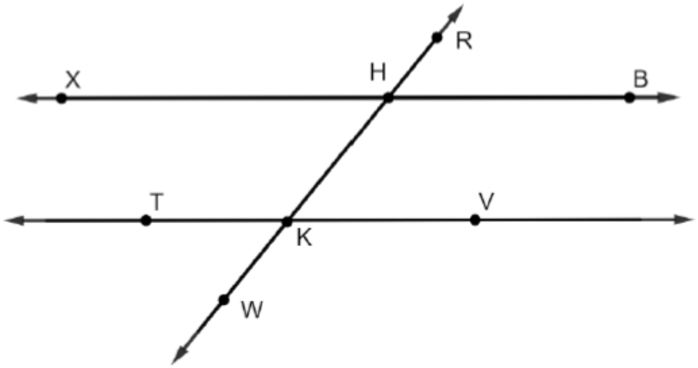
For students who struggle to identify the angle pair relationships, provide the students with two tables for them to draw the parallel lines twice, one with transversal $\overleftrightarrow{MT}$ and one with transversal $\overline{LW}$. Then, in the table with transversal $\overleftrightarrow{MT}$, have the students name all the angles that are congruent to $\angle MLR$ and those congruent to $\angle MLK$. Then, in the table with transversal $\overline{LW}$, have them name those angles congruent to $\angle RLW$ and those congruent to $\angle KLW$. Providing this visual entry-point may help some students to determine which angle relationship to use with each angle pair.

3.7.3 Your Turn!

Component Overview: *Students independently apply angle relationships, postulates, and theorems to solve mathematical and real-world problems.*

3.7.3 Your Turn!

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal with $m\angle XHR = (9x + 21)°$ and $m\angle HKV = (4x + 3)°$.



   a. Determine the value of x.
   $x = 12$

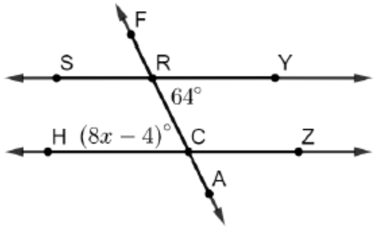
   b. Find the measure of each angle.

$m\angle XHR$ 129°	$m\angle HKV$ 51°
$m\angle TKW$ 51°	$m\angle BHK$ 129°

For students who struggle setting up the equations, have them first highlight or name all the congruent angles to $\angle RHB$ and those congruent to $\angle RHX$. Then, have them state the relationship of angle pairs in the image that are not congruent to each other.

3.7.3 Your Turn! (continued)

2. $\overleftrightarrow{SY} \parallel \overleftrightarrow{HZ}$, $m\angle YRC = 64°$, and $m\angle RCH = (8x - 4)°$.

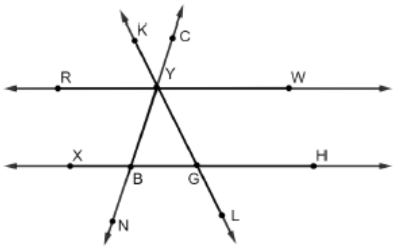


a. Find the value of x.
$x = 8.5$

b. Find the measure of each angle.

$m\angle ZCA$ 64°	$m\angle FRS$ 64°
$m\angle YRF$ 116°	$m\angle HCA$ 116°

3. $\overleftrightarrow{RW} \parallel \overleftrightarrow{XH}$, $m\angle RYK = s°$, $m\angle CYW = m°$, $m\angle GYB = 43°$, $m\angle YBG = 72°$, and $m\angle LGH = (6t + 11)°$.



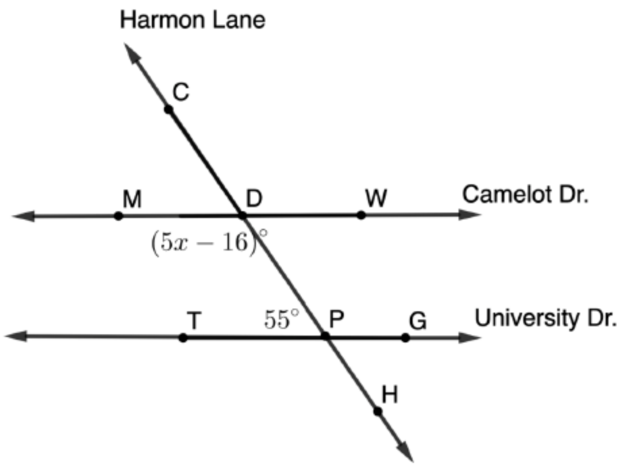
Find the value of each variable. Show your work.

t
$6t + 11 + 72 + 43 = 180$
$t = 9$
m
$m = 72°$
s
$s = (6t + 11)°$
$s = 65°$

If students struggled with question 1, consider working with them individually or in small groups on question 2 to clear up misconceptions and to provide remediated support.

3.7.3 Your Turn! (continued)

4. Harmon Lane, $\overleftrightarrow{CH}$, crosses the streets Camelot Drive, $\overleftrightarrow{MW}$, and University Drive, $\overleftrightarrow{TG}$, as shown. $m\angle TPD = 55°$, and $m\angle MDP = (5x - 16)°$.



Determine the value of x that shows $\overline{MW} \parallel \overline{TG}$.

$x = 28.2$

Challenge students who finish early to write at least two possible assessment questions, one mathematical and one real-world, that involve angle pair relationships formed by two parallel lines and a transversal. Encourage students to be creative, use values and expressions, and provide an answer key. Consider using the questions on a future assessment and award points or credit to any students whose questions are used.

3.7.4. Wrap-Up: Reflection

Component Overview: *Students use a scale to reflect on learning targets related to solving problems involving different angle relationships.*

3.7.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.
Answers will vary.

1. I can use corresponding angles to determine angle measurements.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can use alternate interior angles to determine angle measurements.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can use alternate exterior angles to determine angle measurements.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

4. I can use consecutive interior angles to determine angle measurements.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

5. I can use consecutive exterior angles to determine angle measurements.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

3.8 Angle Relationship Proofs – Part 1

Lesson Introduction

Benchmarks	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Target	• I can prove relationships using parallel lines and their transversals.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How can you prove relationships using parallel lines and their transversals? • What are the similarities and differences between two-column, paragraph, and flow chart proofs?
Lesson Narrative	This lesson uses the angle pair relationships of parallel lines intersected by transversals to complete formal two-column, flow chart, and paragraph proofs. Students will see that some proofs have multiple paths, all of which are correct if there is no contradiction of information and each statement in the proof is built on a given fact, a justification from a given fact, or previous statement. The lesson includes scaffolded collaboration and explicit guided instruction on the process.

Component Summary	3.8.1 Warm-Up (~5 min.)	Would You Rather with two job scenarios *(SE p. 164)*	3.8.3 Guided Instruction (15 min.)	Complete a flow-chart proof and a paragraph proof involving parallel lines and angle relationships. *(SE p. 167)*
	3.8.2 Collaboration (20-25 min.)	Order given steps in proofs involving parallel lines and angle relationships *(SE p. 165)*	3.8.4 Wrap-Up (~5 min.)	Reflect on confidence writing proofs *(SE p. 169)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Parallel Lines • Transversal • Congruent • Supplementary	• N/A	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not realize they can complete proofs in a different order and still be correct.</td><td>Students sometimes lack confidence when writing proofs. Students will need to be continually encouraged to write down everything they know about a given situation and everything they can deduct from what they know. Even if they cannot complete the proof in its entirety, they should be encouraged to start with this as their proof-writing skills continue to develop.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
3.8.2 Collaboration question 1 provides the opportunity for students to discover what information given is contradictory. Before moving on to questions 1a, 1b, and 1c, consider using this routine for students to make observations about what they know is true about the image.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students will be working to complete proofs in various forms. Students who participate in effortful learning, both individually and with others, analyze the problems in a way that makes sense given the task and build perseverance by modifying methods as needed while solving a challenging task. Use the components in this lesson to develop students’ ability to analyze and problem-solve.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing some proper placement of statements and/or reasons for the proof in 3.8.2 Collaboration, question 2. Providing this extra scaffolding can offer a different entry point with students who are struggling with ordering the statements and reasons.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.8.1 Warm-Up: Would You Rather?

Component Overview: *Students compare jobs at two restaurants to determine which one they would prefer to have.*

3.8.1 Warm-Up: Would You Rather?

1. The meals at two restaurants range from $8.00 to $25.00. Servers are paid based on the following.

Restaurant A	Restaurant B
$18 per hour No tipping allowed	$10.50 per hour Tipping encouraged

Would you rather work as a server at Restaurant A or at Restaurant B? Justify your choice.
Answers will vary. I would choose Restaurant A because the meals may be less expensive and require less work with serving tables. I would choose Restaurant B because it could be a place where the meals are more expensive, and I have a good personality to where I can provide great service and be nice to customers and make a lot more in tips.

Think-Pair-Share
Consider implementing the Think-Pair-Share instructional routine for 3.8.1 Warm-Up. Students will make better sense of thinking and reasoning with the opportunity to independently process, then refine answers with a partner, before the teacher elicits more precise responses as a whole group.

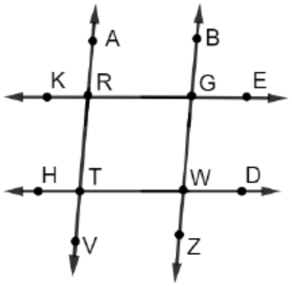
Challenge students to find a rationale that would make both choices equally appealing.

3.8.2 Collaboration: Parallel Lines

Component Overview: *Students examine the given information for a figure to determine if any of the information contradicts itself. Students will then place the steps of a two-column proof in an order that justifies a statement to be proven. Then, the students complete the same proof in an alternative way to show that proofs can follow different correct paths.*

3.8.2 Collaboration: Parallel Lines

1. Contradictory information is given for the figure. $\overleftrightarrow{KE} \parallel \overleftrightarrow{HD}$, $\angle KRA \cong \angle BGE$, $\angle TRG$ is a supplement of $\angle RGW$, and $m\angle DWZ = 87°$.



   a. What piece of information is contradictory?
   Answers will vary. $m\angle DWZ = 87°$

   b. How do you know it is contradictory?
   Answers will vary. If $\angle KRA \cong \angle BGE$ and $\angle TRG$ is a supplement of $\angle RGW$, then both transversals would have to be parallel and intersect the other two parallel lines at $90°$ angles. But, since the $m\angle DWZ = 87°$, this contradicts the angle being equal to $90°$.

   c. Share your contradiction with your partner. Discuss any differences.

   d. Create four different pieces of information about this figure using $\overleftrightarrow{AV} \parallel \overleftrightarrow{BZ}$. One of the pieces of information should be contradictory.
   Answers will vary.

1.	$\angle HTV \cong \angle TWZ$
2.	$\angle RTW$ *is a supplement of* $\angle GWT$
3.	$\angle HTR \cong \angle KRT$ *(contradictory statement)*
4.	$\angle KRT \cong \angle RGW$

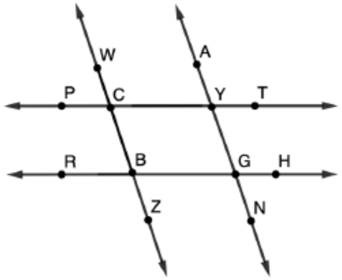
Notice and Wonder
Question 1 provides the opportunity for students to discover what information given is contradictory. Before moving on to questions 1a, 1b, and 1c, consider using this routine to provide the opportunity for students to make observations about what they know is true from the given information, before engaging with the questions.

This collaboration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 3.8.3 Guided Instruction that follows.
Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed collaboration.

e. Trade information with a partner and see if your partner can find the contradiction. Write their contradiction in the table.

Partner's Contradiction	Partner's Initials
Answers will vary.	

2. Terrence was organizing a card sort for the proof, but his cards fell on the floor and are now out of order.



Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle CBG \cong \angle CYG$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$

Statement	Reason
A. $m\angle CBG = m\angle CYG$	K. Definition of Supplementary Angles
B. $\angle CBG$ and $\angle BGY$ are supplementary	L. Substitution Property of Equality
C. $\angle CBG \cong \angle CYG$	M. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
D. $\angle CYG$ and $\angle BGY$ are supplementary	N. Definition of Congruence
E. $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	P. Given
F. $m\angle CBG + m\angle BGY = 180$	R. If two parallel lines are intersected by a transversal, then the consecutive interior angles are supplementary.
G. $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	S. Definition of Supplementary Angles
H. $m\angle CYG + m\angle BGY = 180$	T. Given

Notice 3.8.2 Collaboration, question 1 provides a variety of entry points for learners to engage with the content. Question 1a provides visual cues, question 1b offers space for individual reasoning, question 1c positions students to verbally apply reasoning with their partner, and question 1d allows for students to apply their knowledge of both the process of this problem and their understanding of the material. Some students may have a preference toward one part of the question. Ensure equitable time for all four parts of the question, as each has its benefits for building deeper understanding.

Consider asking the following after question 1e:

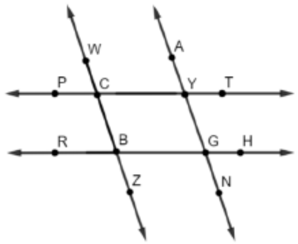
- *Why did you select the information you did for question 1e?*
- *Was it hard coming up with more examples? Why or why not?*
- *Was it hard to write a contradictory statement?*
- *Which statements were easier to write, the correct statements or the contradictory one? Why do you think that is?*

3.8.2 Collaboration: Parallel Lines (continued)

Work with your partner to correctly place the statements and reasons in the correct order by writing the letter.

Statement	Reason
1. *G*	1. *P or T*
2. *C*	2. *P or T*
3. *A*	3. *N*
4. *D*	4. *R*
5. *H*	5. *K or S*
6. *F*	6. *L*
7. *B*	7. *K or S*
8. *E*	8. *M*

3. Work with your partner to complete the proof in a different way.



Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle CBG \cong \angle CYG$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$

Statement	Reason
1. $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	1. Given
2. $\angle CBG \cong \angle WCY$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle CBG \cong \angle CYG$	3. Given
4. $\angle WCY \cong \angle CYG$	4. Transitive Property of Congruence
5. $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	5. *If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*

Consider providing some proper placement of statements and/or reasons for the proof in question 2. Providing this extra scaffolding can offer a different entry point with students who are struggling with ordering the statements and reasons.

For students who may struggle to organize the statements and reasons, consider printing them out on a separate paper to be cut out to provide an alternative method for completing question 2.

While students are working on sorting the statements and reasons for question 2, monitor and make note of different orders the students place them in. Consider stopping students before moving on to question 3 to highlight different ordering, and how more than one order can be correct.

The proof in question 3 uses the same information as question 2. This was done intentionally to show that not only can the same statements be ordered differently, but also different statements in justifying a conclusion from a given may be taken all together. The goal is to show students there is more than one way to prove a relationship. The scaffolds in both question 2 and question 3 are purposeful to lead to two different proofs. There are a different number of steps and different statement and reasons in each.

3.8.3 Guided Instruction: Proofs Involving Angle Relationship

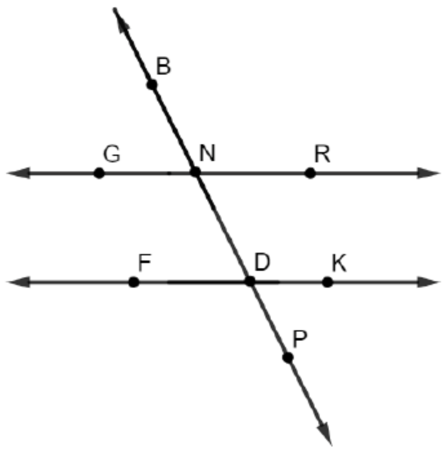
Component Overview: *Students are guided through writing a flowchart proof and a paragraph proof given information related to parallel lines intersected by a transversal.*

3.8.3 Guided Instruction: Proofs Involving Angle Relationship

1. Complete the flowchart proof.

   Given: $\angle RND$ and $\angle FDP$ are supplementary.

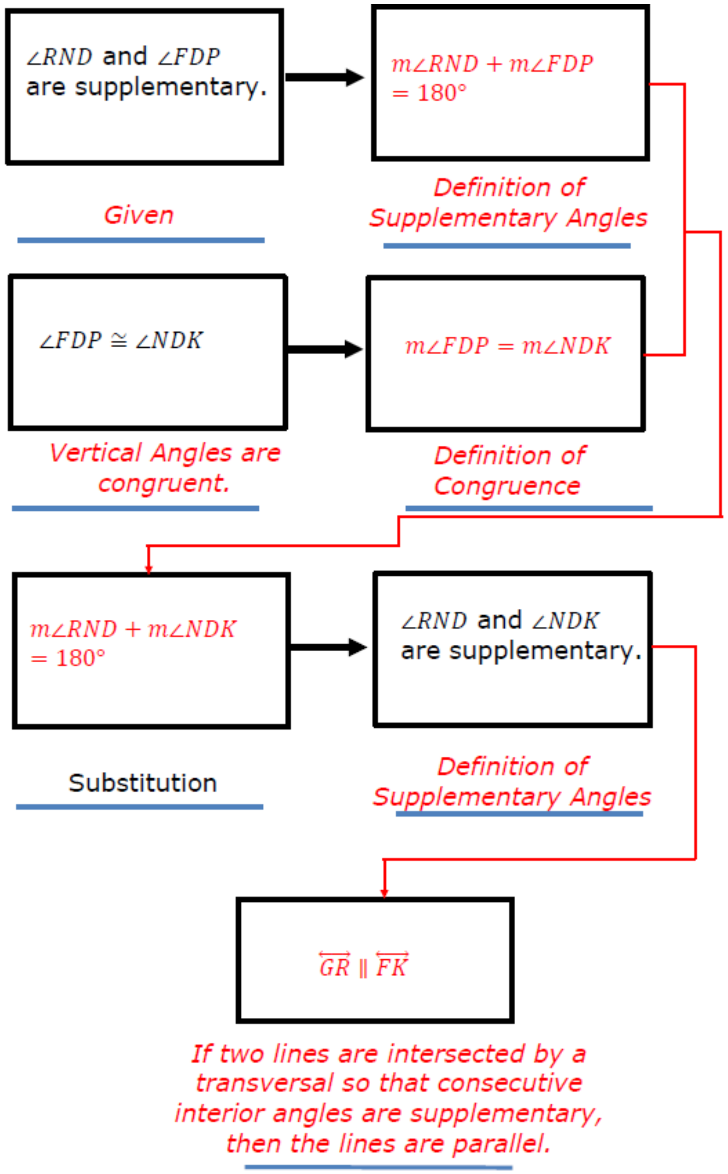
   Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



Students often find flowchart and paragraph proofs more challenging than the two-column proof. This component is designed to be guided so that teachers can help students become more familiar with these forms. Consider using think-aloud and questioning strategies while modeling the type of thinking used in completing these proofs.

Engage students in meaningful dialogue related to the similarities and differences between the different types of proofs. Encourage students to point out the benefit(s) of each form of proof.

3.8.3 Guided Instruction: Proofs Involving Angle Relationship (continued)

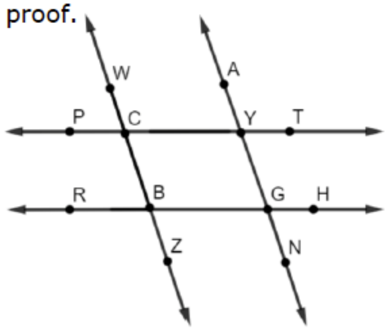


Guide students through the completion of this flowchart proof. Students should fill in the blank boxes with missing statements, and blank lines with missing reasons. Students should also draw arrows showing the connection of the statements, which is one of the benefits of a flow chart proof.

2. Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle PCW \cong \angle HGN$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



It is given that $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$.

Therefore, $\angle RBC \cong \angle PCW$ because if two parallel lines are intersected by a transversal then corresponding angles are congruent.

It is also given that $\angle PCW \cong \angle HGN$. By the Transitive Property of Congruence, it can be written that $\angle RBC \cong \angle HGN$. Therefore, it can be concluded that $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ because if two lines are intersected by a transversal so that alternate exterior angles are congruent then the lines are parallel.

Statements and Reasons	
• $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ • $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ • $\angle PCW \cong \angle HGN$ • $\angle ZBG \cong \angle HGN$ • $\angle GND$ • $\angle FDP$	• Definition of Congruence • Transitive Property of Congruence • Definition of Supplementary Angles • Corresponding angles are congruent • Alternate exterior angles are congruent • Alternate interior angles are congruent

Allow informal observation throughout the first part of this lesson to determine how "guided" this question should be in your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation with students actively working as a whole group.
The bank of statements and reasons provides a support for students, so you could also consider releasing the proof to partners to attempt on their own before going over with the whole class to clarify any questions or misconceptions.

Consider asking the following after completing question 2:
- *If given a choice, why might you select one type of proof to complete over another type?*
- *Will this always be true? Why or why not?*

3.8.4. Wrap-Up: Reflection

Component Overview: *Students will complete a reflection by circling an icon that describes their understanding of each identified aspect of writing proofs.*

3.8.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.
   Answers will vary.

I need help. I got it. I could help someone.

Using *substitution* as a reason in a proof.	
Using the *transitive property of congruence* as a reason in a proof.	
Using the *definition of congruence* as a reason in a proof.	
Understanding what angles are congruent when two lines are parallel.	
Understanding what angles are supplementary when two lines are parallel.	

The information provided from the students here, coupled with what was observed during instruction, can provide valuable data for differentiating instruction. Recording this information and pairing it with assessment data provides insightful feedback for students' learning progressions.

3.9 Angle Relationship Proofs – Part 2

Lesson Introduction

Benchmarks	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Target	• I can prove relationships using parallel lines and their transversals.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How can you prove relationships using parallel lines and their transversals?
Lesson Narrative	Students will complete a Card Sort activity of eight proofs of varying difficulty to practice proving relationships using parallel lines and their transversals.

Component Summary	3.9.1 Warm-Up (~5 min.)	Notice and Wonder about ant hills *(SE p. 170)*	3.9.2 Activity (35-40 min.)	Complete two-column proofs involving parallel lines and transversals by sorting given statements and reasons *(SE p. 170)*
	3.9.3 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 174)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Parallel Lines • Transversal • Congruent • Supplementary	• (Optional) Chart Paper • (Optional) Markers • (Optional) Laminated Proofs and dry erase markers	• The teacher will need to have printed the Statement and Reason Cards, and have them separated into sets for each group. It is recommended to have group sizes of 2-3 students so the sets should be assembled depending on your class size. • If implementing the Take Turns routine, divide the cards up in each set for students to be assigned to each role.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may mix up which cards are statements and which are reasons.</td><td>This lesson centers on a Card Sort activity. Consider using one of the following methods to complete this activity.
Traditional Card Sort
Gallery Walk
Chart Completion
Laminated Proofs
If students struggle completing proofs, consider having a sheet available to hand out where some of the steps have been completed for them.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Take Turns
For the Card Sort in 3.9.2 Activity, set students up to use the Take Turns routine. Decide ahead of time how best to divide the sorting cards between the partners to ensure that both students are participating and one student doesn’t automatically take over the activity.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students who participate in effortful learning, both individually and with others, are able to analyze the problem in a way that makes sense, given the task. They will be able to ask questions that will help them solve the task. This will build perseverance by allowing them to modify methods, as needed while solving a challenging task.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The Card Sort activity offers many opportunities for differentiation. The whole class can work through the proofs in the same way, or different groups can be assigned to work through the activity in different ways, such as small group instruction. Students can also be assigned only some of the proofs instead of completing all of them. Use data from the previous lesson to determine appropriate differentiation to meet the needs of your students.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.9.1 Warm-Up: Notice and Wonder

Component Overview: *Students are provided with a photo taken of ant hills on a brick pathway and are asked what they notice and wonder.*

3.9.1 Warm-Up: Notice and Wonder

A photo of ant hills taken in Gainesville, Florida is shown.

1. What do you notice?
   Answers will vary. The ant hills lie on the creases formed by the straight lines between the bricks.

2. What do you wonder?
   Answers will vary. Do that ant hills have tunnels underneath that connect them and do they run parallel to the creases in the bricks.

3. Write a caption for the photo.
   Answers will vary. Ants Are Great Geometry Students

Do you know someone who is a bricklayer or landscape architect? Consider inviting a guest speaker to join your class virtually or in person. Students can ask the guest speaker questions.
Consider polling the class to share students' creative captions for the photo.

3.9.2 Activity: Card Sort

Component Overview: *3.9.2 Activity positions students to strengthen their knowledge of the relationships between angles and their transversal(s) and skills by completing proofs. They will demonstrate their knowledge of the different relationships as well as their understanding of what can be deduced from given and known information.*

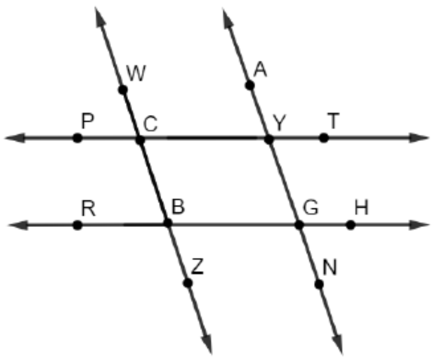
3.9.2 Activity: Card Sort

1. With your group, sort through the statements and reasons for each proof. Then, write each proof in the tables provided.

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$, $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$



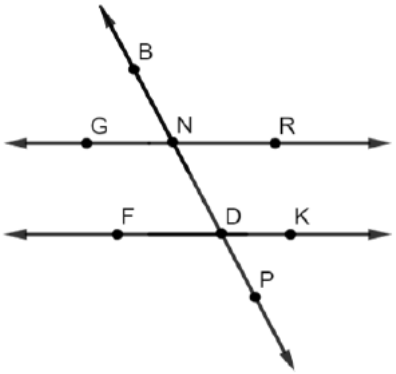
Statement	Reason
$\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	*Given*
$\angle WCY \cong \angle YGH$	*Given*
$\angle WCY \cong \angle AYT$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$\angle AYT \cong \angle YGH$	*Transitive Property of Congruence*
$\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	*If two lines are intersected by a transversal so that the corresponding angles are congruent, then the lines are parallel.*

There are several ways to consider implementing this activity.

Option 1: Traditional Card Sort – suggested group size of 3.
Option 2: The teacher can assign students certain cards, and then have the students do a Gallery Walk. The students can be instructed to do the Card Sort, then write an incomplete proof on chart paper. The students can then circulate in groups to write in the next statement or reason. Then, the group that sorted the cards can check the proof.
Option 3: The empty proofs can be put on chart paper, and students circulate filling in one statement or reason until all the proofs are complete.
Option 4: The teacher copies the blank proofs and laminates them. The students use dry erase markers to complete the proof, and trade with another group to have the proof evaluated.

Consider developing a checklist for your use as you observe students working on the proofs.

3.9.2 Activity: Card Sort (continued)

Proof B	
Given: $\angle GND$ and $\angle PDK$ are supplementary Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	

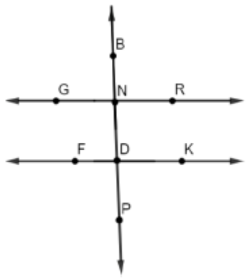
Statement	Reason
$\angle GND$ and $\angle PDK$ are supplementary	Given
$m\angle GND + m\angle PDK = 180°$	Definition of Supplementary Angles
$\angle GND \cong \angle BNR$	Vertical Angles are Congruent
$m\angle GND = m\angle BNR$	Definition of Congruence
$m\angle BNR + m\angle PDK = 180°$	Substitution Property of Equality
$\angle BNR$ and $\angle PDK$ are supplementary	Definition of Supplementary Angles
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	If two lines are intersected by a transversal so that the consecutive exterior angles are supplementary, then the lines are parallel.

Notice and Wonder
For each proof, have students think about the following:

- *What do you notice?*
- *What is the same?*
- *What is different?*
- *What does that make you wonder or think about this proof and how it differs from the first one you completed?*
- *Will this always be true? Why or why not?*

If students struggle with completing these proofs, consider having a sheet available to hand out where some of the steps have been completed for them.

3.9.2 Activity: Card Sort (continued)

Proof C	
Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$ Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	
Statement	**Reason**
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	*Given*
$\angle GNB$ is a right angle	*Definition of Perpendicular Lines*
$m\angle GNB = 90°$	*Definition of Right Angle*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*Given*
$\angle GNB \cong \angle FDN$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$m\angle GNB = m\angle FDN$	*Definition of Congruence*
$m\angle FDN = 90°$	*Substitution Property of Equality*
$\angle FDN$ is a right angle	*Definition of Right Angle*
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	*Definition of Perpendicular Lines*

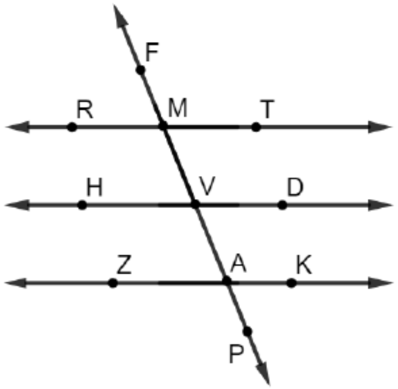
When sharing the correct answers for 3.9.2 Activity, consider having groups present their findings, requiring that they also include their explanation and rationale for their answer choices.

3.9.2 Activity: Card Sort (continued)

Proof D

Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$,
$\angle TMV \cong \angle AVD$

Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$



Statement	Reason
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$	*Given*
$\angle TMV \cong \angle KAP$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$\angle TMV \cong \angle AVD$	*Given*
$\angle KAP \cong \angle AVD$	*Transitive Property of Congruence*
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.*

Consider using some of the time during 3.9.2 Activity to pull small groups of students to provide additional support based on their performance with this component.

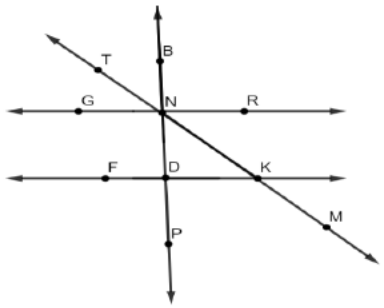
3.9.2 Activity: Card Sort (continued)

For an additional challenge, consider having students write the proof(s) in a different format, such as a flowchart proof or a paragraph proof.

Proof E

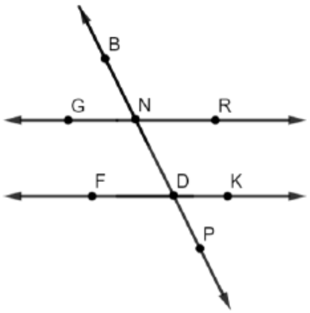
Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

Prove: $\angle DKM$ and $\angle GNT$ are supplementary



Statement	Reason
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	*Given*
$\angle BNR$ *is a right angle*	*Definition of Perpendicular Lines*
$m\angle BNR = 90°$	*Definition of a Right Angle*
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	*Given*
$\angle NDK$ *is a right angle*	*Definition of Perpendicular Lines*
$m\angle NDK = 90°$	*Definition of Right Angle*
$m\angle BNR = m\angle NDK$	*Substitution Property of Equality*
$\angle BNR \cong \angle NDK$	*Definition of Congruence*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.*
$\angle DKM$ *and* $\angle GNT$ *are supplementary*	*If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.*

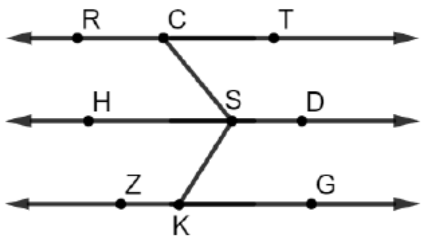
3.9.2 Activity: Card Sort (continued)

Proof F	
Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$ Prove: $\angle GND$ and $\angle PDK$ are supplementary	
Statement	**Reason**
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*Given*
$\angle GND$ *and* $\angle FDN$ *are supplementary*	*If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.*
$m\angle GND + m\angle FDN = 180°$	*Definition of Supplementary*
$\angle FDN \cong \angle PDK$	*Vertical Angles are Congruent*
$m\angle FDN = m\angle PDK$	*Definition of Congruence*
$m\angle GND + m\angle PDK = 180°$	*Substitution Property of Equality*
$\angle GND$ *and* $\angle PDK$ *are supplementary*	*Definition of Supplementary Angles*

3.9.2 Activity: Card Sort (continued)

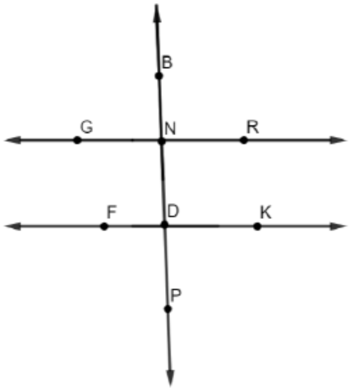
Proof G

Given:
$\angle TCS \cong \angle HSC$,
$\angle HSK \cong \angle SKG$
Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$



Statement	Reason
$\angle TCS \cong \angle HSC$	*Given*
$\overleftrightarrow{RT} \parallel \overleftrightarrow{HD}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*
$\angle HSK \cong \angle SKG$	*Given*
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZG}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$	*Transitive Property of Parallel Lines*

Students may not be clear about the transitive property of parallel lines. The transitive property of parallel lines states that if $\overleftrightarrow{RT}$ is parallel to $\overleftrightarrow{HD}$, and $\overleftrightarrow{HD}$ is parallel to $\overleftrightarrow{ZG}$, then $\overleftrightarrow{RT}$ is parallel to $\overleftrightarrow{ZG}$.
Pause the class or small groups to clarify, as needed.

Proof H	
Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$ Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	
Statement	**Reason**
$\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$	*Given*
$\angle BNR$ *is a right angle.* $\angle NDK$ *is a right angle.*	*Definition of Perpendicular Lines*
$m\angle BNR = 90°$ $m\angle NDK = 90°$	*Definition of a right angle*
$m\angle BNR = m\angle NDK$	*Transitive Property of Equality*
$\angle BNR \cong \angle NDK$	*Definition of Congruence*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the two lines are parallel.*

How is Proof H similar to Proof C? How are they different?

3.9.3 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

3.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about what you've learned about angle relationship proofs. Include things that are easy to remember and common errors to watch out for.
   Answers will vary.